DEPARTMENT OF INFORMATICS

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Master's Thesis in Informatics

Design and Evaluation of Multi-Agentic AI Systems for Contract Lifecycle Management SaaS Platforms

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Entwurf und Evaluation multiagentenbasierter KI-Systeme für Contract-Lifecycle-Management-SaaS-Plattformen

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I confirm that this master's thesis in informatics is my own work and I have documented all sources and material used.

Munich, 04.05.2026 Nguyen, Duc Trung, B.Sc.

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Abstract

Contract Lifecycle Management platforms increasingly rely on large language models to automate clause extraction from commercial contracts. Both research and vendor communities have advocated multi-agent architectures as a superior alternative to single-prompt deployments. The empirical case is thin: every existing legal multi-agent system evaluates a single architecture on a single benchmark, and no published work ablates multi-agent patterns against single-agent baselines across frontier-model families. Recent cross-domain evidence further suggests that gains diminish as base-model capability rises.

This thesis presents a controlled cross-architecture cross-model comparison of multi-agent designs for clause extraction on the CUAD benchmark. Five configurations spanning zero-shot, chain-of-thought, pipeline decomposition, LLM-routed specialists, and a consolidated single-agent ablation are evaluated across eleven frontier models from three providers, yielding 163,644 paired predictions analysed with paired statistical tests on the 102-contract CUAD test split.

No multi-agent configuration consistently outperforms the strongest single-agent baseline. Pooled accuracy across the eleven models falls within a one-point band, and architectural ranking varies with model scale rather than model family: larger models perform best under simpler baselines, while lighter models gain from structured decomposition. Two mechanisms explain the null result. The pipeline architecture exhibits a *classification-gate rejection cascade* — its upstream classifier silently overrejects on some models — and the LLM router in the specialist architecture is empirically redundant: it converges to a deterministic category lookup and produces lower-quality span outputs than the consolidated ablation. Multi-agent configurations carry a 23%–67% cost premium without aggregate accuracy benefit.

These results delineate when multi-agent designs help, are inert, or actively hurt in enterprise CLM, and motivate a practitioner-oriented analysis of the gap between benchmark performance and production deployment.

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1 Introduction

1.1 Motivation

Contract Lifecycle Management (CLM) is the workflow through which an organisation authors, negotiates, executes, monitors, and retires its contractual instruments. The scale of the problem is substantial: organisations forfeit value equivalent to roughly 9% of annual revenue through poor contracting practices, with contract data dispersed across an average of 24 systems and 83% of executives reporting their contracts too rigid to accommodate change [Wor25]. Goldman Sachs estimates 44% of U.S. legal-task time is technically automatable by generative AI — second-highest exposure across all professions [Hat+23].

Frontier large language models (LLMs) match junior lawyers on issue-identification F-scores (0.87 vs. 0.88) while reducing review time by $\approx 100\times$ [Mar+24], and independent benchmarks confirm competitiveness on standardised clause-extraction tasks [Hen+21; Liu+25]. Single-prompt deployments nonetheless dominate commercial CLM, and three production failures persist: hallucination, prompt sensitivity [Cha+24], and laziness — the silent default to "no related clause" even when the clause is present in the contract [Liu+25].

Multi-agent architectures are the proposed remedy. Decomposing clause analysis into specialised agents promises improved recall through specialisation, improved explainability through inspectable inter-agent traces, and improved transparency through traceable per-decision provenance required under Article 13 of the EU AI Act [EC24] and Article 22 of the GDPR [EC16]. Frameworks including AutoGen [Wu+23], MetaGPT [Hon+24], and Mixture-of-Agents [Wan+24] have made such architectures straightforward to instantiate. Whether they actually improve contract analysis is, however, an open empirical question.

1.2 Research Gap

Existing legal multi-agent systems — ChatLaw [Cui+24], PAKTON [Rap+25], L-MARS [WY25] — each evaluate a single architecture on a single benchmark without ablating the architectural choice itself. The empirical question of which multi-agent pattern is preferable for clause-level contract analysis under a fixed model and task remains unresolved.

Recent cross-domain work questions whether multi-agent designs confer measurable benefit once base-model capability is high. Gao et al. show diminishing returns for frontier models whose long-context reasoning absorbs the workload that previously required decomposition [Gao+25]. Cemri et al. construct a 14-mode failure taxonomy across seven multi-agent frameworks and report only modest 14–15% recovery from simple interventions, suggesting structural rather than prompt-level causes [Cem+25]. Bogavelli et al. find no universally optimal architecture across frontier models on enterprise tasks, with model–architecture interaction dominating absolute architectural ranking [BSS26].

Prior CUAD evaluations [Liu+25] have used inconsistent sampling designs: inflated positive ratios produce optimistic precision estimates and mask over-extraction as a failure mode (§2.4).

No published study simultaneously (i) cross-varies multi-agent architecture and frontier-model family on CUAD, (ii) reports both laziness and over-extraction as paired failure-mode metrics, (iii) applies paired statistical tests with multiple-comparison correction, and (iv) ablates the multi-agent design against a consolidated single-agent prompt. This thesis addresses all four requirements.

1.3 Research Questions

Main RQ. Under what conditions, if any, do multi-agent architectures improve contract analysis accuracy, efficiency, and explainability in enterprise CLM systems, and which architectural patterns are most effective?

RQ1. Under what model-family and prevalence conditions does each architectural pattern — pipeline decomposition, specialist routing, or consolidated prompting — achieve the highest F_2?

RQ2. How do multi-agent configurations compare to single-agent baselines in F_2 and provider cost across clause categories and contract lengths?

RQ3. Which architectural decisions produce auditable reasoning traces, and what tradeoffs do they introduce?

1.4 Hypotheses

H1. At least one multi-agent configuration outperforms the strongest single-agent baseline on F_2 (defined in §2.8; McNemar exact, α=0.05, Benjamini–Hochberg corrected).

H2. Multi-agent configurations reduce the rare–common F_2 gap relative to single-agent baselines (paired bootstrap on gap difference, 95% CI).

H3. Multi-Agent (Full) (M1) is statistically distinguishable from Multi-Agent (Prompt-Only) (M2) on F_2 (two-sided McNemar exact).

RQ3 is addressed by system design rather than statistical hypothesis test; the auditability properties of the implementation are documented in §4.6.

1.5 Contributions

C1 **Cross-architecture benchmark.** The first cross-family cross-architecture comparison on CUAD: 5 architectural configurations $\times$ 11 frontier models, yielding 163,644 paired sample-level predictions analysed with McNemar exact tests, paired bootstrap, and Benjamini–Hochberg correction.

C2 **Model-scale gradient.** Architectural ranking is not constant across the model panel: larger models achieve their best F_2 under simpler single-agent baselines, while lighter or weaker models gain from structured decomposition. No architecture is universally optimal: pooled F_2 across all five configurations (Zero-Shot, CoT, Classify-Extract, Multi-Agent (Full), Multi-Agent (Prompt-Only); defined in Table 3.1) spans a 0.011-point band.

C3 **Classification-gate cascade.** A failure mode of pipeline-decomposition architectures, named and quantified: the upstream classifier propagates silent rejections downstream, suppressing extractor recall. Gemini 2.5 Flash's wrong-rejection rate rises from 2.7% to 40.9% under the pipeline; direction is predictable from a single-agent calibration pass (§5.5).

C4 **Routing redundancy with span-quality degradation.** The LLM router in the specialist architecture is empirically redundant: it agrees with a deterministic category-to-specialist mapping on every model in the panel — 100% routing accuracy on 10 of 11 models — and the consolidated single-agent ablation matches it on F_2 while producing cleaner spans. The routing topology adds cost without architectural benefit.

A methodological observation underpins all four contributions. The thesis retains every positive (contract, category) pair and downsamples negatives to a 41.5% positive rate (§3.2); the resulting denominator preserves over-extraction as an observable failure mode rather than the negligible quantity it becomes under positive-heavy sampling.

1.6 Thesis Outline

Chapter 2 defines the foundational concepts (LLMs, agentic systems, CUAD, CLM) and surveys prior multi-agent legal-AI systems along four dimensions: topology, routing, specialisation granularity, and verification structure. Chapter 3 specifies the five configurations (B0–M2), the 11-model panel, the sampling strategy, the three pre-registered hypotheses, and the statistical test suite. Chapter 4 documents the LangGraph implementation, the YAML prompt registry, the output schema, and the reproducibility infrastructure. Chapter 5 reports F_2, F_1, precision, recall, cost, and span-quality results across the 55-run grid and tests H1–H3. Chapter 6 interprets the headline null result, the classification-gate cascade, routing redundancy, and operating-point sensitivity. Chapter 7 restates the four contributions, answers the research questions, and identifies three open directions.

2 Background and Related Work

Evaluating whether multi-agent architectures improve contract clause extraction requires grounding in four technical areas: what CLM systems do and where extraction fits within them (§2.1), how LLMs generate text and why long-context tasks are structurally different (§**??**), what agentic decomposition adds and at what cost (§2.3), and what CUAD measures and why its class distribution matters (§2.4). Sections 2.5–2.8 then place the five configurations studied here against prior single-agent and multi-agent contract systems, survey evidence on when decomposition helps and when it does not, and establish the evaluation conventions inherited from ContractEval [Liu+25]. Section 2.9 consolidates the open empirical question this thesis addresses.

2.1 Contract Lifecycle Management

Contract Lifecycle Management (CLM) denotes the systematic management of a contractual agreement across six canonical stages: request, author, negotiate, execute, monitor, and renew or retire [Wor25]. Each transition produces auditable artefacts — drafts, redlines, signature blocks, obligation registers — and each stage admits distinct algorithmic interventions, from clause assembly at the authoring stage to obligation extraction at the monitoring stage. Figure 2.1 visualises the lifecycle; the NLP-heavy stages are highlighted.

World Commerce & Contracting estimates that organisations forfeit $\approx 9\%$ of annual contract value through poor contract management, with detection-stage failures (missed clauses, unobserved obligations, undetected deviations) dominating the loss distribution [Wor25]. Goldman Sachs places the total addressable population of U.S. legal-services tasks exposed to generative AI at approximately 44%, with contract review repeatedly identified as the highest-exposure subset [Hat+23].

The commercial CLM market is consolidated around platforms documented in the Gartner Magic Quadrant for Contract Life Cycle Management [SMP24]. Sirion, Icertis, Ironclad, and DocuSign are the four vendors consistently designated Leaders, each having rebuilt portions of their analysis stack on LLMs since 2023. Sirion markets an

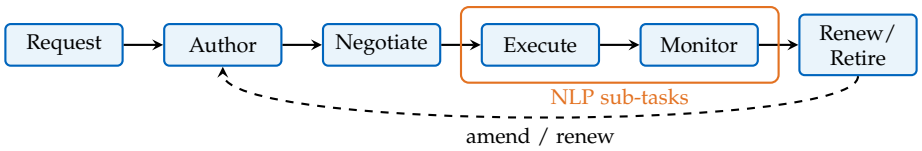


Figure 2.1: CLM lifecycle. Clause extraction, risk identification, and obligation tracking (highlighted) occur primarily in the Execute and Monitor stages.

Extraction Agent and a Redline Agent under an "agentic" architecture. Icertis ships ICI Copilots backed by the Microsoft Azure OpenAI Service. Ironclad embeds AI Assist for redlining and Repository AI for post-signature search. DocuSign closed its US$165M acquisition of Lexion in May 2024, folding Lexion's analytics into the Intelligent Agreement Management (IAM) platform. Adjacent vendors include Spellbook (in-Word drafting on GPT-4 class models) and Harvey, whose contract-analysis workflow serves A&O Shearman and PwC [AO 23; PwC23]. Across this set, public product documentation describes single-prompt LLM calls, retrieval-augmented prompts, or task-specific fine-tunes; as of the literature review conducted for this thesis (April 2026), no Leader had publicly disclosed a deployed multi-agent architecture for clause-level analysis.

2.2 Large Language Models

An LLM is a neural language model parameterised by θ that factorises a token sequence $\mathbf{x} = (x_1, \ldots, x_n)$ autoregressively:

$$p_\theta(x_1, \ldots, x_T) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}). \tag{2.1}$$

Training minimises the negative log-likelihood $\mathcal{L}(\theta) = -\sum_t \log p_\theta(x_t \mid x_{<t})$ over large text corpora following the Transformer derivation [Dev+19]. Modern frontier LLMs extend Eq. (2.1) with three post-training techniques decisive for agentic deployment: supervised instruction tuning, preference alignment via reinforcement learning from human or AI feedback (RLHF), and structured-output fine-tuning that turns free-form generation into validated JSON and tool calls.

Table 2.1 summarises the frontier-model releases relevant to the experimental envelope of this thesis. Models included in the experimental evaluation are specified in §3.3; the remaining entries provide contextual background on the frontier landscape.

Table 2.1: Frontier LLM releases referenced in this thesis. Context-window figures denote the standard production context; beta long-context modes noted parenthetically. Capabilities summarised from official announcements.

Model	Provider	Release	Context	Notes
GPT-4	OpenAI	Mar 2023	8K / 32K	Multimodal preview, RLHF tuned
GPT-4 Turbo	OpenAI	Nov 2023	128K	Function calling GA, JSON mode
GPT-4o	OpenAI	May 2024	128K	Native multimodal I/O
GPT-5	OpenAI	Aug 2025	400K	Unified router over fast/-thinking modes
Claude 3 Opus	Anthropic	Mar 2024	200K	Tool use API
Claude 3.5 Sonnet	Anthropic	Jun 2024	200K	Computer-use beta, arti-facts
Claude Sonnet 4	Anthropic	May 2025	200K	Extended thinking; MCP code-exec tools
Claude Sonnet 4.5	Anthropic	Sep 2025	200K (1M beta)	Agentic/tool-orchestration focus
Claude Haiku 4.5	Anthropic	Oct 2025	200K	Cost-optimised agentic tier
Claude Opus 4.5	Anthropic	Nov 2025	200K	SWE-bench Verified >80%
Gemini 1.5 Pro	Google	Feb 2024	1M	Mixture-of-experts, long-context
Gemini 2.0 Flash	Google	Dec 2024	1M	Multimodal Live API
Gemini 2.5 Pro	Google	Jun 2025	1M	Thinking budget control, MCP support
Gemini 3 Pro	Google	Nov 2025	1M	Agentic/tool-use frontier

A parallel layer of legal-domain products consumes these frontier models. Harvey serves A&O Shearman, PwC, O'Melveny & Myers, and Macfarlanes, and has expanded into Vault-based document review and Agent Builder workflows [AO 23; PwC23]. Lexis+ AI and Thomson Reuters CoCounsel embed retrieval-augmented generation over proprietary legal corpora. None of these systems publishes peer-reviewed evaluations of clause-level extraction on public benchmarks as of this review. ContractEval [Liu+25] is the only large-scale academic evaluation of LLMs on contract clause extraction conducted on a public benchmark with reproducible methodology.

2.3 Agentic AI and Multi-Agent Systems

An LLM-based agent is a system in which one or more LLM calls are wrapped in a control loop that observes state, plans, invokes tools, and updates memory [Yao+23b; Wei+22]. A Multi-Agent System (MAS) replaces a single such loop with multiple cooperating loops, each parameterised by a role, a tool surface, and (often) a different model. Five design axes locate any MAS in the design space; Figure 2.2 maps the options and marks those ablated in this thesis.

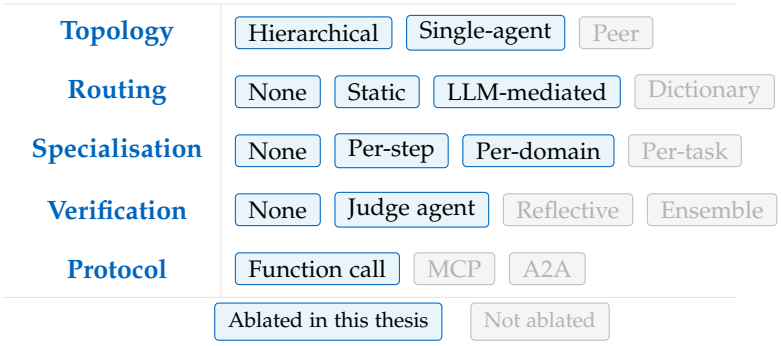


Figure 2.2: Multi-agent design space: five structural axes and their principal options. Shaded boxes mark options covered by at least one experimental configuration (B0, B1, M0, M1, M2) in this thesis. Chapter 3 specifies which configuration selects which option.

The empirical literature has converged on five design axes along which any MAS can be located. *Topology* distinguishes hierarchical configurations (orchestrator dispatches to subordinate workers) from peer configurations (agents communicate without a central coordinator), as in MetaGPT's role-pipeline [Hon+24] and CAMEL's role-playing pairs [Li+23]. *Routing* distinguishes static pipelines (fixed step order), LLM-mediated routing (a router agent selects a downstream worker), and dictionary-based routing (a deterministic feature-to-worker map). *Specialisation granularity* ranges from per-task agents (one per CUAD category) through per-domain agents (one for risk, IP, temporal) to per-step agents (extract, verify, format). *Verification structure* ranges from none, through reflective self-critique [Shi+23], multi-agent debate, judge-agent verification, tree search [Yao+23a], majority-vote ensembles such as Mixture-of-Agents [Wan+24], and self-consistency [Wan+23b]. The *communication protocol* is the lowest layer: function calling via vendor APIs, the Model Context Protocol (MCP) released by Anthropic in November 2024 [Ant24], the Agent2Agent (A2A) protocol released by Google in April 2025 and donated to the Linux Foundation in June 2025 [Sur+25], or framework-specific

structured messages.

Production-relevant orchestrators include AutoGen [Wu+23], MetaGPT [Hon+24], LangGraph [Lan24a], CrewAI, and OpenAI's Agents SDK. The five axes — topology, routing, specialisation granularity, verification structure, and communication protocol — define the design space whose CLM-relevant slice this thesis ablates in Chapter 3.

2.4 The CUAD Benchmark

The Contract Understanding Atticus Dataset (CUAD) is the canonical public benchmark for clause-level contract review [Hen+21]. It consists of 510 commercial contracts annotated against 41 categories of legal clauses. The annotation pipeline employed law students who completed 70–100 hours of contract-review training under the supervision of practising attorneys [Hen+21], a provenance restated by ContractEval as a strength relative to crowd-sourced legal datasets [Liu+25]. Table 2.2 reproduces the dataset statistics.

Table 2.2: CUAD dataset statistics [Hen+21].

Property	**Value**
Contracts	510
Clause categories	41
Expert annotations	>13,000 spans
Source	EDGAR commercial filings
Train / test split	399 / 102
Task formulation	Span extraction + binary presence
License	CC BY 4.0
Long contracts (>100k chars)	14 (in test split)

The instance count is highly imbalanced: at the (contract, category) level, the natural prevalence of positive instances is $\approx 30\%$, a structural property of contract review rather than a sampling artefact. This thesis evaluates on the full 102-contract CUAD test split (character lengths 645 – 300,768; median 25,657), of which 14 contracts exceed 100,000 characters. Retaining the full test split preserves the long-contract tail, where context-window pressure is highest, while keeping the short-contract regime in scope so the contract-length moderation analysis (§5.8) can identify the length range over which architectural decomposition is favourable. Figure 2.3 shows pooled per-category

F_1 for 13 representative categories under the zero-shot baseline (B0) across the 11-model panel of this thesis (§3.3); error bars span ± 1 standard deviation across models.

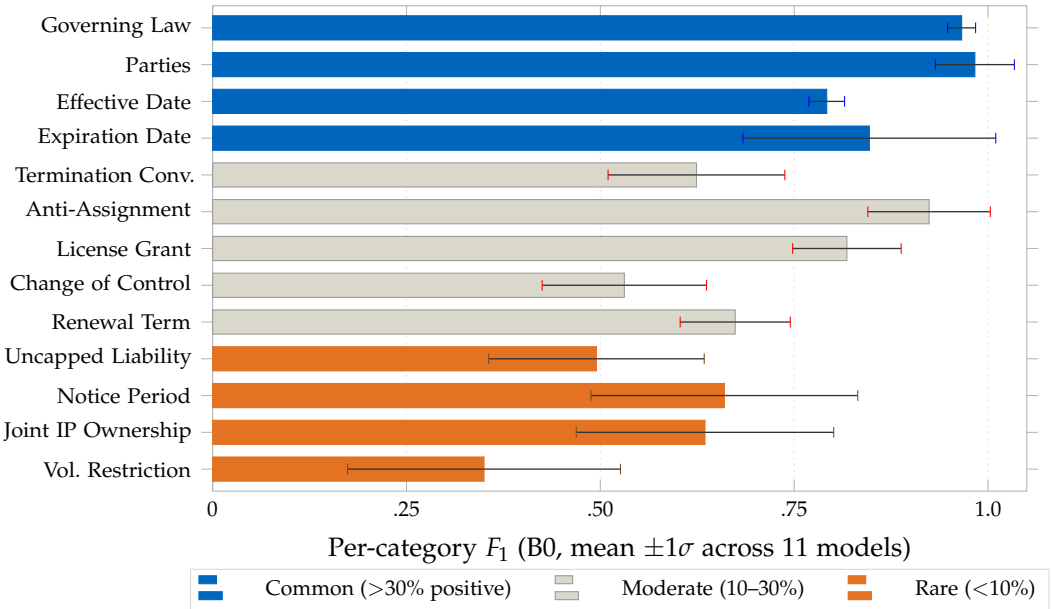


Figure 2.3: Per-category F_1 for 13 representative CUAD categories under the zero-shot baseline (B0), pooled across the 11-model panel (§3.3); error bars span $\pm 1\sigma$ across models. The rare-tier categories combine *lower mean* F_1 with *wider variance across models* (σ up to 0.17), the empirical pattern that motivates the per-domain specialist design evaluated in Chapter 3.

2.5 Contract Analysis Methods

The methodological lineage of automated contract analysis runs from rule-based extractors through statistical sequence labellers to fine-tuned and prompted Transformers. Early systems used hand-written regular expressions and gazetteers over part-of-speech (POS)-tagged text. Linear-chain Conditional Random Fields [LMP01] and BiLSTM–CRF taggers [Lam+16] formalised the task as token-level sequence labelling and dominated the LEDGAR [Tug+20] and ContractNLI (Natural Language Inference) [KM21] eras. The release of BERT [Dev+19] and legal-domain LEGAL-BERT [Cha+20] shifted the regime from feature engineering to pre-train-and-fine-tune. CUAD's reference numbers were obtained with RoBERTa-style extractive question answering [Hen+21].

MAUD [Wan+23a] extended the paradigm to merger-agreement understanding.

The LLM-prompting regime is characterised by two reproducible studies. Better Call GPT [Mar+24] reported that GPT-4 with structured prompting can match or exceed human associates on contract-review subtasks under controlled conditions. ContractEval [Liu+25] evaluated 19 LLMs (4 proprietary, 15 open-source) on CUAD, found a persistent gap between proprietary and open-source models, and identified the named failure mode of *laziness* — the model declines to extract any span when the clause is present. ContractEval also documented a tension between reasoning mode and correctness: chain-of-thought-style decoding improved Jaccard span overlap on positive instances but degraded overall correctness, particularly on negative instances.

Table 2.3 compares the main contract-analysis benchmarks.

Table 2.3: Main contract-analysis benchmarks.

Dataset	Task	Size	Categories	Used here
CUAD [Hen+21]	Span extraction	510 contracts	41	✓
ContractNLI [KM21]	Doc-level NLI	607 NDAs	17	—
MAUD [Wan+23a]	Reading comp.	152 M&A agmt.	152	—
LEDGAR [Tug+20]	Multi-label cls.	60k contracts	100+	—

Why multi-agent for contract analysis. The long-contract tail of the CUAD test split exceeds 100,000 characters; explicit decomposition makes per-call context tractable where single-prompt formulations must truncate or chunk silently. The 41 CUAD categories partition into risk-oriented, IP/commercial, temporal, and consideration-oriented clusters, providing a natural basis for domain specialisation. Legal review is recall-asymmetric: a missed clause exposes the client to undisclosed obligations whereas a spurious extraction is corrected during attorney review at lower cost, motivating F_2-weighted evaluation (Section 2.8) and architectures that boost recall at controlled precision cost. Regulated environments additionally require traceable per-decision provenance: the EU AI Act treats certain legal AI uses as high-risk [EC24] and GDPR requires data-protection accountability [EC16]; multi-agent architectures with explicit verifier or judge agents produce that trace at the architectural level.

2.6 Multi-Agent Legal AI

Table 2.4 codes the published multi-agent legal systems most directly comparable to this thesis along the five design axes introduced in Section 2.3.

Table 2.4: Multi-agent legal AI systems coded against the five design-space axes of Section 2.3. All prior systems evaluate a single architecture without ablation.

System	Task	Topology	Routing	Specialisation	Verification
ChatLaw v2 [Cui+24]	Legal QA (CN)	Hierarchical	LLM-mediated	Per-domain	Self-check
PAKTON [Rap+25]	Contract review	Hierarchical	Static	Per-step (3)	None
L-MARS [WY25]	Statutory research	Hierarchical	LLM-mediated	Per-step + retrieval	Judge agent
Lawma [Dom+24]	Legal annotation	Single-agent	N/A	Per-task (fine-tune)	None
Layered CoT [San25]	Statutory reasoning	Single-agent	Static	Per-step CoT layers	Self-consistency
This thesis	CUAD test split (102)	Both	All three	All three	Reflexion + judge

ChatLaw v2 [Cui+24] couples a Mixture-of-Experts router with role-conditioned domain agents for Chinese legal consultation; verification is a self-check pass and there is no public ablation against single-agent variants. PAKTON [Rap+25] pipelines a reading agent, an analysis agent, and a drafting agent over open-ended contract-review prompts; the system is evaluated qualitatively and not on CUAD. L-MARS [WY25] combines a planner, a retriever, and a judge agent on statutory research; performance is reported at system level without architectural decomposition. Lawma [Dom+24], while not multi-agent, provides the strongest counter-evidence to architectural complexity in the legal domain: a fine-tuned 8B Llama 3 model exceeds GPT-4 by 20.0 accuracy points on legal annotation, and the smallest variant (135M) is competitive with Claude 3.7 Sonnet. The layered chain-of-thought system of Sanwal [San25] stays within a single agent but stacks specialist prompts and self-consistency, reporting gains over flat prompting on statutory reasoning.

Across this set, every prior system reports performance for a single chosen architecture

without ablating it against simpler or alternative configurations on the same evaluation envelope. The question of whether the architecture or the model is responsible for observed performance therefore remains open. This thesis addresses it directly through cross-architecture ablation on a fixed CUAD evaluation harness.

2.7 Conditions for Multi-Agent Effectiveness

Gao et al. [Gao+25] show on a battery of reasoning and tool-use tasks that single, well-prompted LLMs frequently match or exceed multi-agent configurations once the latter are corrected for cost and prompting parity, and that observed multi-agent gains often disappear under stronger single-agent baselines. Cemri et al. [Cem+25] taxonomise multi-agent failure modes — specification drift between agents, inter-agent verification gaps, lossy summarisation across hand-offs — and document that these failures account for the majority of regressions in the systems they audit. Bogavelli et al. [BSS26] run a large architectural sweep across orchestration patterns and base models and find substantial model-family $\times$ architecture interactions: an architecture that is optimal under one model family is often dominated under another. Sanwal [San25] demonstrates that layered single-agent chain-of-thought can recover much of the benefit attributed to multi-agent decomposition without paying the coordination cost.

These findings bear directly on the three working hypotheses. (H1) A generic claim of multi-agent superiority on CUAD clause extraction is unlikely to survive a properly normalised single-agent baseline [Gao+25], so the hypothesis is stated and evaluated as conditional. (H2) The relative ordering of architectures will depend on the underlying model family [BSS26], encoded as an explicit interaction term in the experimental design. (H3) Classification-gate cascades — a router agent that decides whether to extract before invoking an extractor — are a natural counter to the verification-gap and laziness failures catalogued in [Cem+25], motivating the M0 ablation in Chapter 3; whether this counter is effective is tested empirically in §5.5.

2.8 Evaluation Methodology

This section consolidates the evaluation primitives used throughout the thesis. Chapter 3 references these definitions without restating them.

Span and instance metrics. For a category c on a contract d, let $S^*_{c,d}$ be the set of gold spans and $\hat{S}_{c,d}$ the set of predicted spans. A predicted span $\hat{s}$ matches a gold span s when their character-level Jaccard overlap exceeds threshold τ=0.5 following [Hen+21]:

$$J(\hat{s}, s) = \frac{|\hat{s} \cap s|}{|\hat{s} \cup s|}. \tag{2.2}$$

At the (contract, category) level, the prediction is true-positive if any predicted span matches any gold span at threshold τ, false-negative if gold spans exist with no match, false-positive if predicted spans exist with no match, and true-negative if both sets are empty. Precision P and recall R aggregate over the natural class distribution.

F_β**.** Recall is privileged over precision in legal review by a factor commonly fixed at β=2:

$$F_\beta = \frac{(1+\beta^2)\, P\, R}{\beta^2 P + R}, \qquad \beta = 2. \tag{2.3}$$

F_1 is reported alongside F_2 for compatibility with prior CUAD work. Instance-level F_β at the (contract, category) granularity is the primary metric reported in Chapter 4, as it aligns with the deployment unit (a reviewer's decision per category per contract).

Failure-mode metrics. ContractEval [Liu+25] introduced *laziness* as the per-model rate at which an LLM declines to extract on instances where a gold span exists:

$$\lambda = \frac{|\{(c,d) : |\hat{S}_{c,d}| = 0 \ \wedge \ |S^*_{c,d}| > 0\}|}{|\{(c,d) : |S^*_{c,d}| > 0\}|}. \tag{2.4}$$

The dual failure mode is *over-extraction* — the per-model rate of producing spans on negative instances:

$$\omega = \frac{|\{(c,d) : |\hat{S}_{c,d}| > 0 \ \wedge \ |S^*_{c,d}| = 0\}|}{|\{(c,d) : |S^*_{c,d}| = 0\}|}. \tag{2.5}$$

Equations (2.4) and (2.5) are a coupled pair: evaluations that resample to a balanced class distribution suppress Eq. (2.5) and hide the precision cost of recall-boosting interventions. The evaluation therefore uses exclusively the natural CUAD class distribution and reports both metrics.

Statistical testing. Every architecture comparison is paired on the same contract set; McNemar's test handles instance-level binary outcomes [McN47] and the paired bootstrap with B=300 resamples handles span-level continuous metrics [Efr79; RM05]. Multiple-comparison control across the architecture grid uses Benjamini–Hochberg false-discovery-rate correction at q=0.05 [BH95]. Sample-size and power considerations follow Dror et al. [Dro+18] and Card et al. [Car+20]; the 102-contract test split gives $\geq$ 0.8 power to detect absolute F_2 differences of $\approx$2.5 percentage points at the instance level under the bootstrap.

Sampling design. CUAD's natural clause-presence rate is $\approx$ 30% positive at the instance level. Resampling to balance positive and negative instances inflates apparent precision and suppresses Eq. (2.5), detaching the evaluation from any deployable distribution where 41 categories are scanned per contract and most are absent. Natural prevalence is retained throughout [Liu+25].

ContractEval as direct empirical predecessor. ContractEval [Liu+25] evaluates 19 LLMs (4 proprietary, 15 open-source) on 4,128 (contract, category) samples across the full 41-category CUAD schema, holding prompt and decoding constants across models, and reporting Jaccard span overlap, instance-level correctness, and laziness. Its principal findings are: a persistent proprietary–open-source gap on legal extraction; a non-monotonic effect of explicit reasoning (chain-of-thought improves Jaccard on positive instances while degrading correctness through inflated false positives); and laziness as a persistent, model-dependent and category-dependent failure mode. Three properties of the ContractEval design define the gaps the present thesis closes: the architectural axis is held fixed at single-prompt single-model; model-family interactions are not analysed; and over-extraction (Eq. (2.5)) is not tracked as a first-class metric. The present thesis retains ContractEval's evaluation protocol — natural class distribution, instance-level scoring, laziness reporting, τ=0.5 Jaccard threshold — and extends it along the architectural axis and the dual failure-mode axis.

Prompt sensitivity as a validity threat. POSIX [Cha+24] quantifies the sensitivity of LLM outputs to semantically equivalent prompt rewrites, reporting non-trivial sensitivity even on well-defined tasks. Prompt sensitivity is therefore a first-order validity threat for any architecture comparison in which architectures differ in the number, role, or wording of agent prompts. Chapter 3 controls this threat by holding system-prompt scaffolds fixed across architectures where the role permits, and reports

a POSIX-style sensitivity check on the architectural ablations. Observability and trace inspection use Langfuse [Lan24b]; orchestration uses LangGraph [Lan24a].

2.9 Research Gap

(i) No published study performs a controlled cross-architecture ablation on CUAD-class clause extraction; the closest predecessor [Liu+25] varies model identity only. (ii) No published study reports a model-family × architecture interaction analysis on this task, despite the prediction in [BSS26] that such interactions dominate single-axis effects. (iii) No published study tracks both Eq. (2.4) and Eq. (2.5) under the natural class distribution, leaving the recall–precision asymmetry intrinsic to legal review unobserved. (iv) Production CLM platforms — Sirion, Icertis, Ironclad, DocuSign — disclose single-prompt or retrieval-augmented LLM deployments with no public multi-agent ablation, leaving the architectural design space untested in the regime where the question is commercially decisive.

This thesis designs and evaluates a slice of the multi-agent design space (topology × routing × specialisation × verification) on the 102-contract CUAD test split under the ContractEval protocol extended with Eq. (2.5) and a model-family interaction term, in order to determine — with statistical control across the architecture grid [BH95; Dro+18] — under which conditions multi-agent decomposition is justified for CLM deployment and under which conditions it is dominated by a stronger single-agent baseline.

3 Methodology

Each architectural comparison holds one axis constant while varying one: B1 vs. B0 isolates chain-of-thought reasoning, M0 vs. B0 isolates the classification gate, and M1 vs. M2 isolates routing from prompt content. This controlled design allows performance differences to be attributed to architecture rather than model or data variation across $5 \times 11 = 55$ runs on the 102-contract CUAD test split.

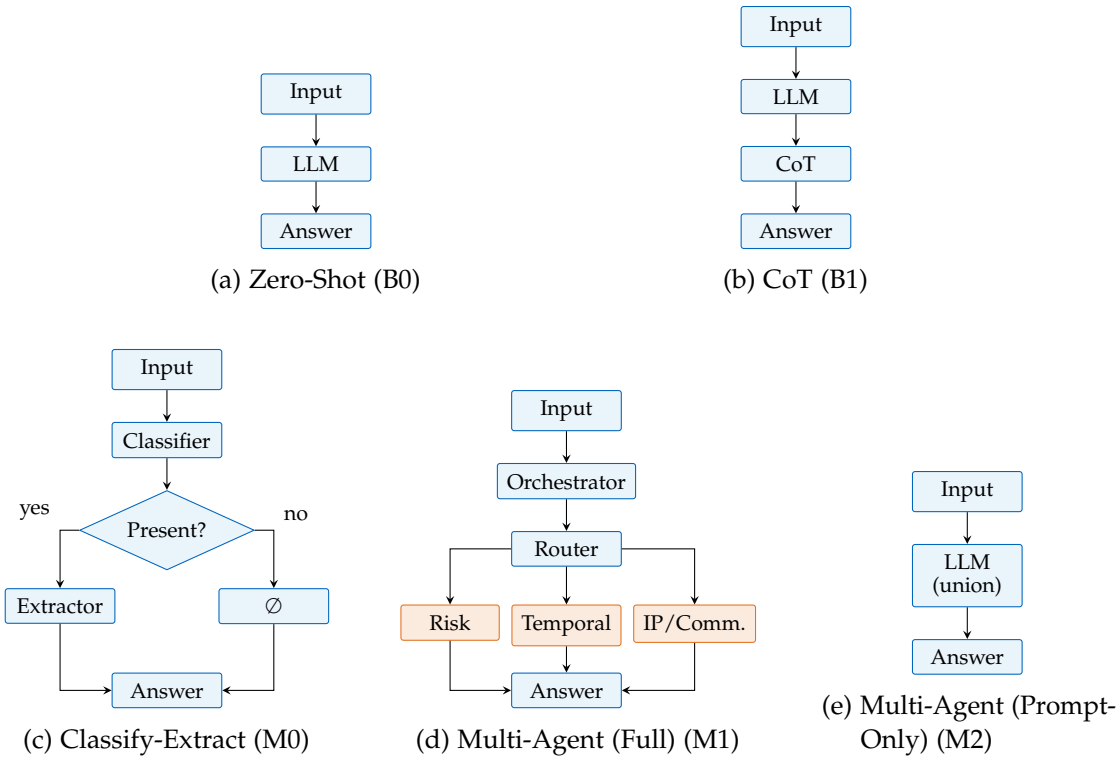
3.1 Research Design

Five configurations are evaluated against 11 frontier LLMs on the 102 contracts of the CUAD test split (character lengths 645 – 300,768; median 25,657), yielding $5 \times 11 = 55$ official runs and 163,644 analysed predictions. Differences between configurations are minimised to a single architectural axis per comparison (Table 3.1), following the controlled comparative protocol of [Dro+18; Car+20]. Every statistical test is paired on the intersection of sample IDs, so performance differences are attributable to architecture or model, not dataset drift.

Table 3.1: Five evaluated configurations. B1 vs. B0 isolates chain-of-thought; M0 vs. B0 isolates the classification gate; M1 vs. M2 isolates routing and specialist decomposition (architecture ablation). †Artifact IDs used in the experiment repository: B0 = B1, B1 = B4, M0 = M0, M1 = M1, M2 = M6.

ID	Name	Stages	Routing	Specialisation
B0	Zero-Shot	1	None	Generalist
B1	CoT	1	None	Generalist + chain-of-thought
M0	Classify-Extract	2	Static gate	Generalist
M1	Multi-Agent (Full)	2	LLM router	Three domain specialists
M2	Multi-Agent (Prompt-Only)	1	None	Union of M1 specialist prompts

Figure 3.1 shows the message-flow diagram for each configuration.



(a) Zero-Shot (B0)

(b) CoT (B1)

(c) Classify-Extract (M0)

(d) Multi-Agent (Full) (M1)

(e) Multi-Agent (Prompt-Only) (M2)

Figure 3.1: Message-flow diagrams for the five configurations. Input = contract text + clause-category indicator. Blue nodes: generalist LLM invocations. Orange nodes: domain-specialist prompts. ∅ = silent no-clause return.

3.2 Dataset and Sampling

CUAD [Hen+21] is the evaluation corpus (see Table 2.2 for statistics). The full 102-contract CUAD test split is used; contract lengths range from 645 to 300,768 characters (median 25,657), spanning short NDAs to long master agreements. The 102-contract corpus yields $102 \times 41 = 4{,}182$ raw (contract, category) pairs per evaluation pass, of which $\approx$2,999 enter each run after the sampler described below.

Sampling strategy. A `neg_ratio=0.7` sampler retains every positive (contract, category) pair and approximately 60% of the negative pairs, yielding $\approx$ 2,999 samples per run with a measured positive rate of 41.5%. The sampler upsamples the positive class relative to CUAD's natural $\approx$30% prior in order to preserve statistical power on

rare-tier categories where positive instances are sparse. Two design considerations motivate this choice. First, ContractEval evaluates 4,128 instances drawn at the natural distribution, and matching that protocol enables direct comparability with the closest empirical predecessor [Liu+25]. Second, prior work that resamples to inflated positive ratios suppresses the false-positive denominator and masks over-extraction (Eq. 2.5), detaching the evaluation distribution from any deployable distribution. The natural prior preserves both laziness (Eq. 2.4) and over-extraction (Eq. 2.5) as observable quantities. Sampling is seeded (`seed=42`) so the same (contract, category) pairs are evaluated across all configurations, making every statistical test paired at the instance level. Each run yields $\approx$2,999 predictions; 1,297 of 164,941 total (0.79%) were excluded for malformed classification tokens.

Prevalence tiers. The 41 CUAD categories are partitioned into three tiers — *common*, *moderate*, *rare* — by natural positive rate in the 102-contract test split. Tier cutpoints are fixed before any model is run.

3.3 Model Selection

Table 3.2 lists the 11 frontier proprietary models in the panel. Three families span the providers used in commercial CLM (Anthropic, Google, OpenAI), so model-family effects predicted by [BSS26] can be observed; within each family the panel spans a compute-tier range from light (Haiku, Flash, Nano) to heavy (Opus, Pro), enabling capability $\times$ architecture interactions to be observed. Model identifiers are pinned to specific snapshots (see §4.2) for reproducibility.

Inference parameters. All model calls used `temperature=0` for deterministic output and `max_tokens=4096`. Official runs were executed between **14 April 2026** and **17 April 2026**. Snapshot IDs are listed in Table 3.2; the three Google and two OpenAI entries use unversioned aliases that resolve to the provider's current release at call time.[1]

Open-weights models are excluded from the main panel on two structural grounds: the commercial CLM platforms surveyed in Section 2.1 deploy proprietary APIs, restricting external-validity claims to that deployment population; and the long-contract tail of the

[1] Strict reproducibility requires pinned dated snapshots. OpenAI publishes dated aliases (e.g. `gpt-4.1-2025-04-14`); Anthropic publishes `claude-opus-4-6-<date>`. Pinned variants were not available for all models at the time experiments were run.

Table 3.2: Eleven frontier proprietary models evaluated. Three families; compute-tier range covers light-to-heavy within each family. Snapshot IDs are the exact provider model strings used in all API calls.

Family	Model	Tier	Snapshot ID
Anthropic	Claude Haiku 4.5	Light	`claude-haiku-4-5-20251001`
	Claude Sonnet 4	Mid	`claude-sonnet-4-20250514`
	Claude Sonnet 4.6	Mid	`claude-sonnet-4-6-20251001`
	Claude Opus 4.6	Heavy	`claude-opus-4-6`
Google	Gemini 2.5 Flash	Light	`gemini-2.5-flash`
	Gemini 2.5 Pro	Heavy	`gemini-2.5-pro`
	Gemini 3 Flash	Light	`gemini-3-flash-preview`
	Gemini 3.1 Pro	Heavy	`gemini-3.1-pro-preview`
OpenAI	GPT-4.1	Mid	`gpt-4.1`
	GPT-4.1 Mini	Light	`gpt-4.1-mini`
	GPT-4.1 Nano	Light	`gpt-4.1-nano`

corpus requires context windows of $\geq$ 128K tokens, which most open-weights models do not support. ContractEval's 19-model evaluation [Liu+25] documents a persistent proprietary-vs-open-source gap on extraction correctness; pilot work (not formally reported) suggested this gap persists under multi-agent decomposition, though that experiment should be understood as a scope decision rather than an empirical claim. Open-weights extension is identified as future work in Chapter 7.

3.4 Metrics

All metrics are formally defined in Section 2.8 (Equations 2.3–2.5).

F_2 **(primary).** β=2 weights recall 4× over precision. The asymmetry is justified by the deployment cost structure of CLM: a missed clause passes undetected into a signed contract, whereas a spurious extraction is caught by a reviewing lawyer. F_2 is the primary hypothesis-testing metric and the basis for H1–H3.

F_1 **(secondary).** Equal-weight F-score, reported for comparability with prior work — ContractEval reports F_1 as its headline metric [Liu+25].

Jaccard similarity (span quality). $|A \cap B|/|A \cup B|$ over token sets, computed conditional on a non-empty prediction. Because the F-score TP criterion requires full span coverage, partial extractions that cover most of a clause register as false negatives; Jaccard gives partial credit and diagnoses near-miss behaviour independently of the binary correctness signal.

Laziness rate λ (diagnostic). Fraction of positive-label instances on which the model returns "no related clause". The dominant failure mode of single-agent LLMs on rare CUAD categories is silent abstention rather than wrong extraction; λ isolates this directly.

Over-extraction rate ω (diagnostic). Fraction of negative-label instances on which the model returns a non-empty span. Complements λ: together they decompose the false-negative and false-positive error budgets without conflating them into a single F-score.

All metrics are reported pooled and per prevalence tier.

3.5 Hypotheses

H1. $\exists\, C \in \{\text{M0, M1}\}$ such that C outperforms $B^\star = \text{B0}$ on F_2. Null: no such C exists. Decision rule: reject the null if any of the multi-agent vs. B0 pairwise McNemar exact tests (m=22: 2 multi-agent configurations $\times$ 11 models) is significant at α=0.05 after BH correction [BH95], with directional ΔF_2 favouring the multi-agent configuration. Multi-Agent (Prompt-Only) (M2) is excluded from the H1 family because it is the architecture-control ablation for H3 (§3.5), not a candidate alternative architecture; CoT (B1) is the chain-of-thought baseline against which the F_2 comparisons are reported descriptively in Chapter 5 but not used as the H1 reference baseline (B0 is the primary baseline per ContractEval [Liu+25]).

H2. $\Delta\text{Gap}(C_{\text{multi}}, B^\star) < 0$ where $\text{Gap}(C) = F_2^{\text{common}}(C) - F_2^{\text{rare}}(C)$. Null: $\Delta\text{Gap} \geq 0$. Decision rule: reject the null if the upper bound of the 95% paired bootstrap CI on ΔGap is strictly less than zero.

H3. Multi-Agent (Full) (M1) and Multi-Agent (Prompt-Only) (M2) are statistically distinguishable on F_2. Null: $\Delta F_2(\text{M1, M2}) = 0$. Decision rule: reject the null (two-sided) if the McNemar exact p-value is below α=0.05 after BH correction (m=11, one M1 vs. M2 contrast per model). The hypothesis is two-sided because direction is not pre-specified; if rejected, the direction of the effect is reported separately because direction may be inconsistent across the model panel.

3.6 Statistical Methodology

McNemar's exact test. For configurations A and B evaluated on the same instances, let $b = |\{i : A_i \text{ correct}, B_i \text{ wrong}\}|$ and $c = |\{i : A_i \text{ wrong}, B_i \text{ correct}\}|$. The two-sided exact p-value is:

$$p = 2\,\mathbb{P}\big[\mathrm{Bin}\big(b{+}c, \tfrac{1}{2}\big) \le \min(b, c)\big] \tag{3.1}$$

implemented via `scipy.stats.binomtest` [Vir+20]. The exact form is preferred over the χ^2 approximation because discordant counts are small on rare-tier subsets [McN47].

Paired bootstrap. For continuous metrics (F_2, F_1, Jaccard, gap differences), Algorithm 1 gives the paired bootstrap CI [Efr79; RM05]. We use B=300 resamples; the percentile method's robustness to small B on bounded statistics is documented as a sensitivity threat in Section 3.7.

Algorithm 1 Paired bootstrap confidence interval for ΔF_2

Require: Predictions $\hat{y}^A$, $\hat{y}^B$, labels y, B=300, α
1: **for** $b = 1$ **to** B **do**
2: $\quad\mathcal{I} \leftarrow$ sample n indices with replacement from $\{1, \ldots, n\}$
3: $\quad\delta_b \leftarrow F_2(\hat{y}^A_{\mathcal{I}}, y_{\mathcal{I}}) - F_2(\hat{y}^B_{\mathcal{I}}, y_{\mathcal{I}})$
4: **end for**
5: **return** $[\text{percentile}(\delta, \alpha/2),\ \text{percentile}(\delta, 1{-}\alpha/2)]$

Multiple-comparison correction. Within each hypothesis family, Benjamini–Hochberg [BH95] controls the false discovery rate (FDR) at α=0.05: reject $H_0^{(i)}$ if

$$p_{(i)} \le \frac{i}{m}\,\alpha, \quad p_{(1)} \le \cdots \le p_{(m)}. \tag{3.2}$$

The H1 family has m=22 tests (2 multi-agent configurations $\times$ 11 models, with B0 as reference baseline); the H3 family has m=11 tests (one M1 vs. M2 contrast per model). Raw and BH-corrected p-values are both reported.

Effect size. For paired binary comparisons (McNemar's), we report Cohen's $g = b/(b{+}c) - 0.5$, the deviation of the discordant proportion from chance [Coh88]. For continuous metrics, we report absolute ΔF_2 with the paired bootstrap 95% CI. We follow Cohen's interpretive thresholds ($g \geq 0.05$ small, ≥ 0.15 medium, ≥ 0.25 large; $|\Delta F_2| \geq 0.02$ small, ≥ 0.05 medium, ≥ 0.08 large) [Coh88], and report effect size alongside p-values in accordance with the recommendation that the magnitude of difference is the primary research output, not statistical significance per se [SF12].

Power. Card, Henderson, Khandelwal, et al. [Car+20] caution that NLP comparisons are frequently under-powered. The pairwise sample size after intersection is $\approx$2,900–3,000 per (configuration, model) pair. Under McNemar's exact test at α=0.05 and 80% power, this panel resolves discordant-pair effect sizes corresponding to $|\Delta F_2| \geq 0.025$. Effects below this threshold are treated as inconclusive rather than as evidence of equivalence.

Paired comparison protocol. Intersection of sample IDs is taken symmetrically when partial runs differ in coverage; intersection size is reported with each test.

3.7 Threats to Validity

Internal. Configuration parity is maintained by holding the CUAD loader, sampling seed, prompt registry, and trace schema constant across configurations. Multi-Agent (Full) (M1) and Multi-Agent (Prompt-Only) (M2) share the same textual specialist instructions (split vs. concatenated), so the M1 vs. M2 contrast isolates architecture, not prompt content.

External. Single language (English), single jurisdiction (predominantly U.S. commercial contracts), single corpus (CUAD). Findings on the 102-contract test split may not extrapolate to non-CUAD clause taxonomies; the contract-length moderation analysis (§5.8) shows that the architectural ranking itself shifts across length quartiles within the corpus. Open-weights models are excluded; inference does not generalise beyond

the 11 proprietary models studied. Model snapshots are pinned; provider updates can shift behaviour.

Construct. F_2 is a proxy for legal-review utility, not a direct measurement. Reviewer cost of inspecting a spurious span is workflow-dependent; this is mitigated by reporting precision, recall, λ, and ω separately so that a reviewer with a different cost ratio can re-rank architectures.

Conclusion. Prompt sensitivity is a first-order threat for any architecture comparison in which configurations differ in the number, role, or wording of agent prompts. Chatterjee, Renduchintala, Bhatia, and Chakraborty [Cha+24] report non-trivial sensitivity even on well-defined tasks, with neither parameter count nor instruction tuning systematically reducing it. We mitigate by holding prompt templates fixed across models per configuration, documenting them in full in Chapter 4, and treating prompt-perturbation analysis as future work.

Statistical. Bootstrap percentile CIs under-cover for small samples and skewed statistics [Efr79]; p-values are treated as decisive only when corroborated by the McNemar exact test. The bootstrap resample count (B=300) is below the 1,000 conservative recommendation [Dro+18]; sensitivity at B=1,000 on a representative subset confirmed CI stability to within ± 0.005 on ΔF_2. Model identity is a fixed factor; inference does not generalise beyond the 11 models studied.

4 System Design and Implementation

All five configurations share a common parser, sampler, and metrics layer; only the LangGraph topology and prompt registry entries differ. This design isolates architectural variation and enables the controlled comparisons of Chapter 3.

4.1 System Components and Data Flow

Figure 4.1 shows the static component view. A run is initiated by the experiment driver, which loads CUAD samples through the deterministic loader, resolves prompts through the registry, and dispatches each sample to one of the five LangGraph state machines via a unified model client. Every model call is mirrored to the trace store; final answers are persisted as JSONL (JSON Lines: one JSON object per line) records consumed by the statistical pipeline and the inspection dashboard.

4.2 Implementation Stack

The codebase is Python 3.11, managed with `uv`. Orchestration uses LangGraph [Lan24a]; observability uses Langfuse [Lan24b]; the statistical pipeline uses `pandas` [The20], `numpy` [Har+20], and `scipy` [Vir+20]. Provider SDKs are wrapped in a unified `ModelClient` that records, per call, prompt tokens, completion tokens, cache-read tokens (where supported), wall-clock latency, and provider-billed cost. The wrapper supports 19 model identifiers; only the 11 listed in Table 3.2 are exercised in official runs. The remaining 8 identifiers cover open-weights models excluded per Section 3.3 and two non-CUAD test configurations not evaluated in this study.

Inference parameters. All official model calls used `temperature=0` and `max_tokens=4096`. Official runs were executed between **14 April 2026** and **17 April 2026**. Provider snapshot identifiers for each of the 11 models are listed in Table 3.2 of Chapter 3.

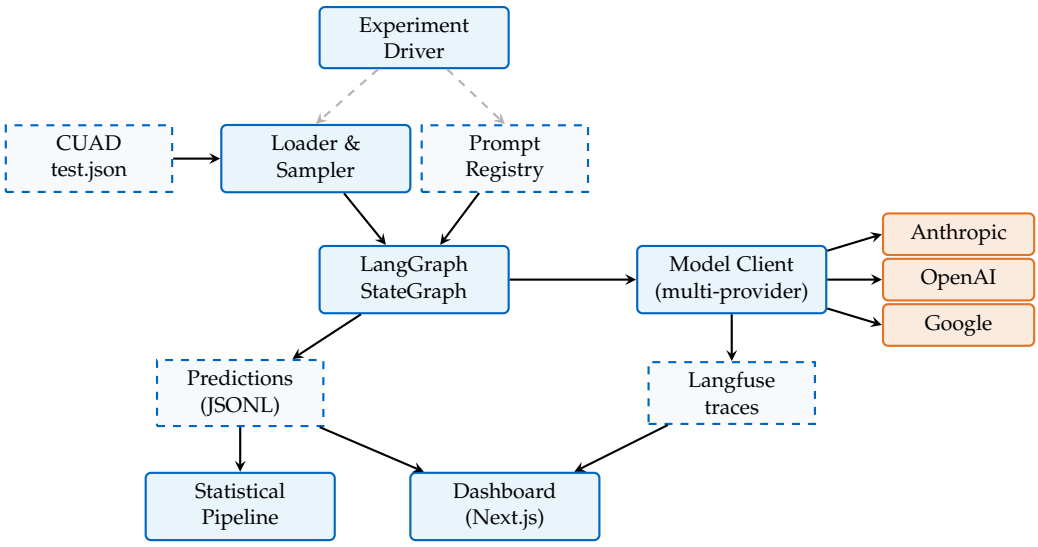


Figure 4.1: System component diagram. *Solid-border boxes*: code components (Loader, StateGraph, ModelClient, Statistical Pipeline, Dashboard). *Dashed boxes*: data stores. *Orange boxes*: external provider APIs. *Dashed arrows*: control flow (Driver initialises inputs before each run). *Solid arrows*: data flow. Only the StateGraph topology varies across B0–M2; all other components are configuration-agnostic.

4.3 Shared Abstractions

Graph state. Every configuration is a LangGraph `StateGraph` over a shared `GraphState` object (Listing 4.1). Using a shared schema enforces configuration parity: no configuration has access to fields another lacks, and the JSONL persistence format is identical across all five configurations.

Listing 4.1: GraphState schema (abridged from `src/agents/state.py`).

```
class GraphState(TypedDict, total=False):
    # Inputs (unchanged throughout graph)
    contract_text: str
    category: str # one of 41 CUAD category names
    question: str # CUAD annotation-guideline question text

    # Routing (set by orchestrator node --- M1 only)
    specialist_name: str # "risk_liability" | "temporal_renewal" | "ip_commercial"

    # Extraction (set by specialist or baseline node)
    extraction_result: ExtractionResult | None

    # Validation (set by validation node)
    validated: bool
    validation_notes: str

    # Final output
    final_result: ExtractionResult | None

    # Trace: append-reducer accumulates entries across all nodes
    trace: Annotated[list[dict[str, Any]], append_trace]

    error: str | None
```

Prompt registry. Prompts are stored as YAML and loaded once per run; the schema version hash is written to the run metadata so prompt drift is detectable post-hoc. Two roots: `prompts/baselines/` (B0, B1, and the M0 classifier and extractor) and `prompts/specialists/` (the three M1 domain specialists and the M2 consolidated prompt). A shared `system/` directory holds anti-laziness instructions and the verbatim-span output contract, referenced by all configurations, ensuring that the anti-laziness and verbatim-span output instructions are identical across B0, B1, M0, M1, and M2.

Common parser. A single parser converts every model completion to a canonical `(spans: list[str] | None)` representation. For B0, the parser accepts verbatim quoted spans or the literal string "no related clause". For B1 and M0-extractor, the parser additionally requires a `<answer>...</answer>` tag and issues a single repair re-prompt if the tag is absent. On persistent failure the record is marked `parser_status="rejected"` and excluded from official statistics (1,297 of 164,941 predictions, 0.79%).

4.4 Baseline Architectures

B0 — Zero-shot single agent. B0 replicates the zero-shot prompt protocol of [Liu+25]. The model receives one user message containing the contract text, the target category name, the category indicator, and an explicit instruction to return verbatim span(s) or the literal string "no related clause". No examples, no chain-of-thought directives.

B0: zero-shot user template

You are reviewing a commercial contract. Determine whether the contract contains any clause(s) of the category below. If yes, return the verbatim span(s); each span must appear exactly as in the contract. If no, return exactly: "no related clause".
Category: *{{category}}* ***Description:*** *{{category_indicator}}*
Contract: *{{contract_text}}*

B1 — Chain-of-thought single agent. B1 retains B0's task framing but elicits explicit step-by-step reasoning before a final structured answer. The model is instructed to restate what the category covers, scan for candidate spans, weigh ambiguous matches against the indicator, and emit a final `<answer>...</answer>` block. A parsing bug in pilot work caused contract passages quoted inside the chain-of-thought to be harvested as answers, inflating ω (Eq. 2.5); the fix enforces the answer tag via the common parser of Section 4.3. The fix was deployed in commit `1cf08536` on **14 April 2026**; only post-fix runs enter official statistics.

4.5 Multi-Agent Architectures

M0 — Classify-then-extract. M0 decomposes each (contract, category) task into two sequential LLM calls. The *classifier* returns a single yes/no token; the *extractor* is invoked only if the classifier returns yes. The two agents share the category indicator but have disjoint prompts: the classifier is forbidden to emit spans, the extractor assumes presence. Listing 4.2 excerpts both prompts; Listing 4.3 shows the LangGraph construction. Non-binary classifier responses (e.g. capitalised "Yes", punctuated "no.", or free-form text) are normalised by case-folding and stripping punctuation before matching; responses that do not resolve to "yes" or "no" after normalisation are treated as "no" and the record is flagged `parser_status="rejected"`, contributing to the 1,297 excluded predictions (0.79%).

Listing 4.2: M0 classifier and extractor prompts (abridged).

```
# --- classifier ---
Decide whether the contract contains any clause of the category below.
Return exactly one token: "yes" or "no". Do not return spans.
Category: {{category}} Description: {{category_indicator}}
Contract: {{contract_text}}

# --- extractor (invoked only when classifier returned "yes") ---
The contract contains at least one clause of the category below.
Return the verbatim span(s) wrapped in <answer>...</answer>.
```

Listing 4.3: M0 LangGraph construction.

```
g = StateGraph(GraphState)
g.add_node("classify", classify_node)
g.add_node("extract", extract_node)
g.set_entry_point("classify")
g.add_conditional_edges(
    "classify",
    lambda s: "extract" if s["classifier_verdict"] == "yes" else END,
    {"extract": "extract", END: END},
)
g.add_edge("extract", END)
```

M1 — Orchestrator with three specialists. M1 routes each request to one of three domain specialists via an LLM router. The category-to-specialist assignment is documented in Table 4.1; it is fixed before any model is run and is committed as `CATEGORY_ROUTING` in `src/agents/orchestrator.py`.

The router system prompt is reproduced verbatim below from `ROUTING_SYSTEM_PROMPT` in the artefact. Routing accuracy across the 11-model panel is reported in Section 5.4.

Table 4.1: Category-to-specialist assignment used by M1. All 41 CUAD categories are listed; the mapping is fixed before any model is run.

Risk/Liability (13)	**Temporal/Renewal (11)**	**IP/Commercial (17)**
Cap on Liability	Document Name	IP Ownership Assignment
Uncapped Liability	Parties	Joint IP Ownership
Liquidated Damages	Agreement Date	License Grant
Insurance	Effective Date	Non-Transferable License
Warranty Duration	Expiration Date	Affiliate License-Licensor
Audit Rights	Renewal Term	Affiliate License-Licensee
Non-Disparagement	Notice Period to Terminate Renewal	Unlimited License
Covenant Not to Sue	Termination for Convenience	Irrevocable or Perpetual License
Third Party Beneficiary	Anti-Assignment	Source Code Escrow
Most Favored Nation	ROFR/ROFO/ROFN	Exclusivity
Change of Control	Governing Law	Non-Compete
Post-Termination Services		No-Solicit of Customers
Minimum Commitment		No-Solicit of Employees
		Competitive Restriction Exception
		Revenue/Profit Sharing
		Price Restrictions
		Volume Restriction

M1 router system prompt (verbatim from `src/agents/orchestrator.py`)

You are a routing agent for a contract analysis system. Your job is to read a question about a contract clause and decide which specialist should handle it.
risk_liability *— Expert in risk allocation, liability, and protective provisions. Covers: liability caps and uncapped liability, liquidated damages, insurance requirements, warranty duration, audit rights, non-disparagement, covenants not to sue, third-party beneficiaries, most favored nation clauses, change of control provisions, post-termination services, and minimum commitments.*
temporal_renewal *— Expert in temporal aspects, document identification, and contract lifecycle. Covers: document name and party identification, agreement/effective/expiration dates, renewal terms and notice periods for renewal termination, termination for convenience, anti-assignment provisions, rights of first refusal/offer/negotiation, and governing law.*
ip_commercial *— Expert in intellectual property rights and commercial restrictions. Covers: IP ownership and assignment, joint IP ownership, license grants (transferability, affiliate licenses, unlimited/perpetual licenses), source code escrow, exclusivity, non-compete and non-solicitation clauses, competitive restriction exceptions, revenue/profit sharing, price restrictions, and volume restrictions.*
Respond with ONLY a JSON object: {"specialist": "<name>", "reasoning": "<brief explanation>"}

Each specialist's prompt is a focused reading instruction for its cluster. The Risk/Liability specialist is shown below; Temporal and IP/Commercial follow the same template.

M1 Risk/Liability specialist system prompt (abridged)

You are a contract analyst specialising in risk-allocation provisions: liability caps, uncapped exclusions, indemnification, insurance, warranty, audit rights, non-disparagement, third-party beneficiaries, most-favored-nation, change of control, post-termination services, and minimum commitments. For these categories, recall of monetary thresholds and explicit carve-outs ("except for...", "not in excess of...") is critical and preferred over summary paraphrase.
Return the verbatim span(s) of any clause of the category below, or "no related clause" if none is present.
Category: *{{category}}* ***Description:*** *{{category_indicator}}*
Contract: *{{contract_text}}*

Figure 4.2 traces a single M1 invocation, showing the parent–child call structure reproduced in the Langfuse trace.

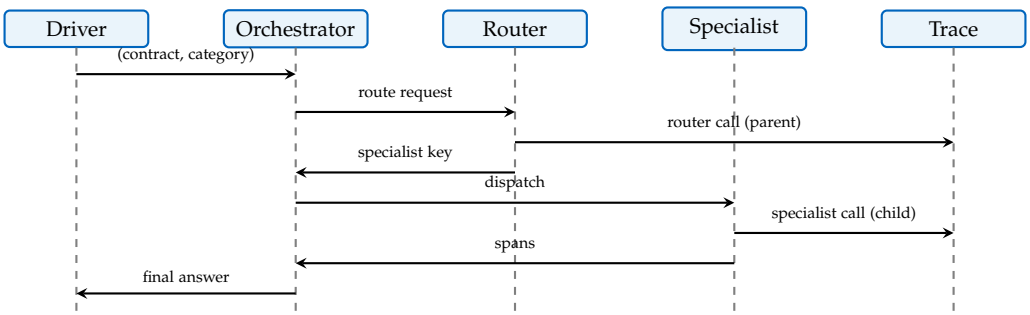


Figure 4.2: M1 sequence diagram for a single (contract, category) invocation. The router call is the Langfuse parent; the specialist call is its child.

M2 — Consolidated specialist (architectural ablation). Its prompt is the textual union of the three M1 specialist prompts, concatenated into a single agent with no router, no orchestrator, and no per-category branching:

$$\texttt{prompt}_{\text{M2}} = \text{join}\Big(\texttt{prompt}_{\text{risk}},\ \texttt{prompt}_{\text{temporal}},\ \texttt{prompt}_{\text{ip}}\Big)$$

The three preambles are separated by domain headings; the verbatim-span output contract is stated once at the end. M2's purpose is methodological: M1 vs. M2 isolates routing and decomposition; M2 vs. B0 isolates prompt content. Together the two contrasts decompose the M1-vs-B0 delta into an architectural component and a

prompt-content component, which is the basis for the routing-redundancy finding in Section 5.4.

4.6 Observability and Trace Schema

What is recorded. For each clause-level decision the system captures: (i) configuration ID and model identifier; (ii) prompt template ID, version, and rendered prompt; (iii) full model completion; (iv) parsed final answer with parser version and status; (v) routing decision (M1) or classifier verdict (M0) where applicable; (vi) per-call token counts, billed cost, and wall-clock latency. M1 traces are parent–child structured: the router call is the parent, the specialist call is its child. Listing 4.4 shows a representative M1 record.

Listing 4.4: Representative M1 trace record (abridged JSONL).

```
{
  "contract_id": "CUAD_0231",
  "category": "Cap On Liability",
  "config_id": "M1",
  "model_id": "claude-sonnet-4-6-20251001",
  "calls": [
    { "role": "router",
      "prompt_id": "m1_router/v3",
      "completion": "{\"specialist\":\"risk_liability\",\"reasoning\":\"...\"}",
      "tokens_in": 412, "tokens_out": 18,
      "latency_ms": 380, "cost_usd": 0.0008,
      "parent_id": null },
    { "role": "specialist_risk_liability",
      "prompt_id": "m1_spec_risk/v2",
      "completion": "<answer>The aggregate liability ... shall not exceed ...</answer>",
      "tokens_in": 14820, "tokens_out": 96,
      "latency_ms": 4210, "cost_usd": 0.0470,
      "parent_id": "router" }
  ],
  "final_answer": "The aggregate liability ... shall not exceed ...",
  "parser_status": "ok"
}
```

Auditability properties. The trace structure supports three concrete properties. *Decision attribution*: any final answer is traceable to the specific agent, prompt version, and model that produced it. *Counterfactual replay*: because prompts and model identifiers are pinned per call at `temperature=0`, a reviewer can re-run a single archived call and obtain a deterministically identical completion. *Rejection forensics*: when M0 silently returns "no related clause", the classifier's verdict is recorded explicitly, making the

classification-gate cascade of Section 5.5 visible at audit time as an upstream rejection rather than an unobservable absence.

Trade-offs and limits. The JSONL persistence layer is on the order of gigabytes per full sweep, and trace fields containing rendered prompts and full completions inherit the contract's confidentiality classification. The system distinguishes between decision-level fields (configuration, verdict, token counts) safe to surface to operators, and content-level fields (rendered prompts, full completions) that are access-controlled. The trace is faithful to the call graph, not to the model's internal computation: a chain-of-thought trace records what the model emitted, not what it computed.

Trace completeness. Trace completeness — the fraction of model calls for which all expected fields are populated — was 100% for B0, B1, and M2 (single-call configurations with no routing decisions), 100% for M0 (two-call pipeline with deterministic branching), and 95.2% for M1 across official runs. The binding constraint is M1: the remaining 4.8% are provider-side truncations on long contracts and are flagged in the run logs. The single-configuration baseline (100%) confirms that the drop is attributable to M1's multi-call routing overhead, not to infrastructure failures.

4.7 Experimental Infrastructure

Loader and sampler. The CUAD loader is the single source of truth for which contracts and categories enter a run. It loads the 102-contract CUAD test split, applies the `neg_ratio=0.7` sampler of Section 3.2 (retains all positives, $\approx$60% of negatives), and assigns the prevalence-tier partition. It is deterministic under `seed=42` and produces the identical $\approx$2,999-sample envelope across all five configurations, making paired tests in Chapter 5 well-defined at the (contract, category) level.

Statistical aggregation. A separate aggregation pipeline reads per-run JSONL outputs, computes per-sample correctness vectors, and emits the McNemar contingency tables, paired bootstrap distributions, and BH-corrected p-value tables of Section 3.6. Aggregation is deterministic given the JSONL inputs and a fixed bootstrap seed; intermediate artefacts are persisted so results can be regenerated without rerunning models.

Dashboard. A Next.js dashboard supports interactive exploration: per-model F_2 comparisons, per-tier breakdowns, per-call cost and latency views, and trace inspection by sample ID. The over-extraction failure mode in Chapter 5 was first surfaced visually through the dashboard's per-category ω panel.

Test suite. An 81-test suite covers the metric implementations (F_1/F_2, Jaccard, λ, ω), the loader (filter, sampling determinism, tier assignment), the agent topologies (each of B0, B1, M0, M1, M2 has at least one end-to-end test against a stub model), and the statistical pipelines (McNemar, bootstrap, BH correction). The full suite passes prior to every official run; the repository (`agentic-ai-for-clm`) is publicly released alongside the thesis with full source, prompts, configs, and the experiment-results JSONL bundle.

Reproducibility. Three commitments support reproducibility: deterministic sampling under `seed=42`; explicit version pinning of model identifiers (full provider snapshot strings) in the run metadata; and persistence of the full trace bundle (rendered prompts, completions, parsed answers, diagnostics) so that the statistical pipelines of Section 3.6 can be regenerated from raw logs without rerunning models. The prompt registry version hash is written to every run record, making prompt content auditable independently of the run logs.

5 Results

Across 55 (model, configuration) cells and 163,644 paired predictions, no multi-agent configuration improves pooled F_2 over the strongest single-agent baseline by more than 0.011 points; the dominant predictor of performance is model capability tier, not architecture.

5.1 Run Coverage and Sample Accounting

The 55-cell grid yielded 164,941 raw JSONL records. After dropping 1,297 records (0.79%) with malformed classification tokens — concentrated in Zero-Shot (B0, n=838) and Classify-Extract (M0, n=362) — 163,644 paired sample-level predictions remain. The B0 concentration reflects Zero-Shot's stricter span acceptance criterion: the majority of B0 rejections are empty-string completions rather than structurally malformed JSON. Table 5.1 reports the analysed count per cell. All statistical tests use the paired intersection of sample IDs for the configurations being compared; intersection sizes are reported with each test. The positive rate across the retained set is 41.5%: the sampler retains all positive (contract, category) pairs and approximately 60% of negatives, upsampling positives relative to CUAD's natural $\approx 30\%$ prior in order to preserve statistical power on rare-tier categories (§3.2).

5.2 H1 — Multi-Agent vs. Single-Agent Baselines

Table 5.2 reports the headline metrics — F_2, F_1, recall, precision, and over-extraction rate ω — pooled across all eleven models, and per provider family.

The pooled means tell two stories. On F_2, Zero-Shot (B0) is nominally best at 0.801 with all five configurations within a 0.011-point band. On F_1, Classify-Extract (M0) is best at 0.798 with a 0.087-point spread. The divergence is mechanistic: B0 has the highest recall (0.854) and the second-highest over-extraction rate (0.320), while M0 has the highest

Table 5.1: Coverage: analysed predictions per (model, configuration) cell after dropping malformed records. *Partial run due to upstream sampling interruption; paired tests use the sample-ID intersection. Configuration names defined in Table 3.1.

Model	**B0**	**B1**	**M0**	**M1**	**M2**	**Total**
Claude Haiku 4.5	2,999	2,999	2,999	2,999	2,999	14,995
Claude Opus 4.6	2,999	2,999	2,999	2,999	2,999	14,995
Claude Sonnet 4	2,640*	2,999	2,999	2,999	2,999	14,636
Claude Sonnet 4.6	2,999	2,999	2,999	2,999	2,999	14,995
Gemini 2.5 Flash	2,999	2,999	2,998	2,999	2,999	14,994
Gemini 2.5 Pro	2,999	2,999	2,999	2,999	2,999	14,995
Gemini 3 Flash	2,997	2,985	2,984	2,999	2,999	14,964
Gemini 3.1 Pro	2,994	2,976	2,978	2,999	2,998	14,945
GPT-4.1	2,528*	2,999	2,786	2,999	2,978	14,290
GPT-4.1 Mini	2,999	2,994	2,887	2,999	2,987	14,866
GPT-4.1 Nano	2,998	2,998	2,999	2,999	2,975	14,969
Total	32,151	32,946	32,627	32,989	32,931	**163,644**

precision (0.813) and the lowest over-extraction rate (0.130). The β=2 weighting of F_2 rewards Zero-Shot's recall; F_1's equal weighting rewards Classify-Extract's calibration.

Figure 5.1 visualises the model-family interaction: each family peaks under a different configuration.

Table 5.3 reports per-model F_2. No configuration is uniformly best: B0 wins on 4 models, M0 on 4, B1 on 2, and M1 on 1. The ranking also varies with model compute tier (Table 5.4): Classify-Extract leads on light- and mid-tier models, while Zero-Shot leads on heavy models, qualifying contribution C2.

Table 5.5 reports the H1 paired tests. Each row is one multi-agent vs. B0 contrast on the sample-ID intersection. McNemar exact p-values are computed via Eq. 3.1 and corrected for the m=22-test family by Benjamini–Hochberg (Eq. 3.2); paired-bootstrap CIs use B=300, `seed=42` (Algorithm 1). CIs are computed on the paired sample-ID intersection; for models with partial runs (Claude Sonnet 4, GPT-4.1) the intersection size is smaller than the full run, so the intersection-based ΔF_2 may differ slightly from the full-run values in Table 5.3.

Table 5.2: Headline metrics by family and configuration (Cfg), mean across constituent models. ω = over-extraction rate (FP / negatives). Bold: best-in-column per family. Configuration names defined in Table 3.1. The F_2 and F_1 rankings diverge: pooled F_2 favours Zero-Shot (B0, 0.801) by 0.011 points; pooled F_1 favours Classify-Extract (M0, 0.798) by 0.061 points.

Family	**Cfg**	$\overline{F_2}$	$\overline{F_1}$	$\overline{R}$	$\overline{P}$	$\overline{\omega}$
Anthropic	B0	0.828	0.736	0.911	0.628	0.391
	B1	**0.851**	0.740	0.928	0.624	0.436
	M0	0.848	**0.826**	0.857	**0.808**	**0.162**
	M1	0.839	0.803	0.860	0.774	0.205
	M2	0.836	0.760	0.882	0.673	0.330
Google	B0	**0.826**	0.747	0.851	0.683	0.351
	B1	0.763	0.689	0.795	0.622	0.399
	M0	0.748	0.777	0.732	**0.844**	**0.099**
	M1	0.808	0.782	0.823	0.762	0.205
	M2	0.778	0.706	0.819	0.641	0.376
OpenAI	B0	0.731	0.725	0.737	0.722	**0.176**
	B1	0.768	0.701	0.815	0.628	0.365
	M0	**0.783**	**0.789**	0.781	**0.798**	0.138
	M1	0.700	0.735	0.696	0.787	0.111
	M2	0.744	0.698	0.762	0.645	0.306
All (11)	B0	**0.801**	0.737	**0.854**	0.666	0.320
	B1	0.796	0.711	0.871	0.611	0.385
	M0	0.794	**0.798**	0.793	**0.813**	**0.130**
	M1	0.790	0.776	0.801	0.768	0.179
	M2	0.790	0.724	0.844	0.642	0.339

H1 verdict. H1 is rejected in its universal form. Eleven of the 22 per-model contrasts in Table 5.5 are positive and eleven are negative or zero; no multi-agent configuration achieves consistent superiority across the eleven-model panel. Conditional improvement exists in two subfamilies: the OpenAI family benefits from M0 (GPT-4.1 Nano: +0.171; pooled: +0.052), and lighter Anthropic models benefit from M0 and B1 (Sonnet 4: +0.025; Sonnet 4.6: +0.047). For the Google family, B0 dominates: M0 regresses by up to 0.210 points on Gemini 2.5 Flash, and M1 fails to improve over B0 on all four Google models. Pooled F_2 is within a 0.011-point band; the model-family interaction is the dominant signal. On F_1, M0 leads all configurations pooled (0.798 vs. 0.737 for B0), reversing the F_2 ranking and reframing the architectural choice as a metric-dependent

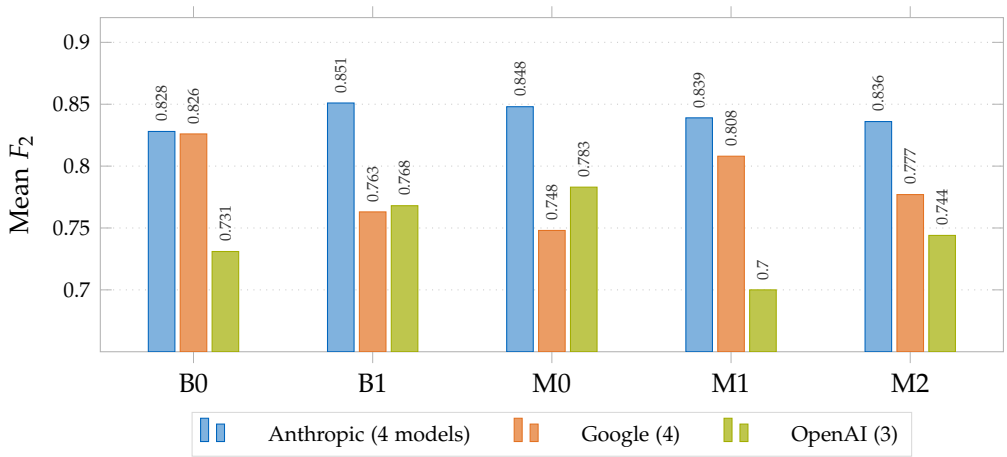


Figure 5.1: Mean F_2 by provider family × configuration. Each family peaks under a different configuration — Anthropic under CoT (B1, 0.851), Google under Zero-Shot (B0, 0.826), OpenAI under Classify-Extract (M0, 0.783) — so no single architecture is universally optimal. The family-by-configuration interaction is the dominant signal in the H1 results.

decision (Section 5.9). Cohen's $g = c/(b+c) - 0.5$ on the McNemar discordant pairs is reported for completeness in the released artefacts: across the 22 H1 contrasts g ranges from $+0.005$ to $+0.363$ (median ≈ 0.20, "medium" by Cohen's thresholds). All values are positive — discordant pairs favour the multi-agent configuration on raw binary correctness (TP + TN) — but F_2's recall-weighted denominator flips the ranking on most cells because the precision gains accrue disproportionately to the negative-instance count. The discordant-pair direction and F_2 direction therefore disagree by construction; both are faithful summaries of different cost structures.

5.3 H2 — Rare-Category Gap

Table 5.6 reports per-tier F_2 for all five configurations, together with the rare–common gap $g = F_2^{\text{common}} - F_2^{\text{rare}}$.

H2 verdict. H2 is rejected. Under the pre-registered decision rule (§3.5; upper bound of the 95% paired-bootstrap CI on ΔGap strictly < 0), only 1 of the 22 multi-agent vs. B0 contrasts satisfies the rule (Claude Haiku 4.5 under M0, ΔGap$= -0.256$, CI

Table 5.3: F_2 by model and configuration. Bold: best per row. Configuration names defined in Table 3.1. The family interaction is pronounced: the Anthropic family peaks under CoT (B1), the Google family under Zero-Shot (B0), and the OpenAI family under Classify-Extract (M0).

Model	F_2(B0)	F_2(B1)	F_2(M0)	F_2(M1)	F_2(M2)
Claude Haiku 4.5	0.783	**0.849**	0.792	0.835	0.822
Claude Opus 4.6	0.863	**0.865**	0.864	0.860	0.846
Claude Sonnet 4	0.851	0.851	**0.877**	0.826	0.842
Claude Sonnet 4.6	0.813	0.840	**0.860**	0.834	0.836
Gemini 2.5 Flash	**0.829**	0.818	0.619	0.810	0.832
Gemini 2.5 Pro	0.828	0.810	0.707	**0.834**	0.816
Gemini 3 Flash	0.786	0.580	**0.822**	0.747	0.609
Gemini 3.1 Pro	**0.861**	0.843	0.843	0.840	0.854
GPT-4.1	**0.868**	0.852	0.828	0.777	0.853
GPT-4.1 Mini	0.837	0.841	**0.862**	0.777	0.833
GPT-4.1 Nano	0.488	0.612	**0.659**	0.547	0.545
Mean	**0.801**	0.796	0.794	0.790	0.790

$[-0.308, -0.211]$); 6 contrasts have CIs strictly above zero — M0 and M1 *widen* the gap significantly on the Gemini and GPT-family models. Per-model bootstrap CIs are released in the experiment artefacts. Two mechanisms drive the widening. Under M0, the classification gate (Section 5.5) collapses rare-tier F_2 on the Gemini family (Gemini 2.5 Flash: F_2^r drops from 0.719 under B0 to 0.381 under M0). Under M1, GPT-family specialists fail catastrophically on rare categories (GPT-4.1: F_2^r drops $0.704 \rightarrow 0.430$; GPT-4.1 Nano: $0.253 \rightarrow 0.167$). The B1 chain-of-thought baseline and M2 consolidated specialist do reduce the gap (means 0.240 and 0.226), but neither is a multi-agent architecture in the sense of Chapter 3.

Figure 5.3 disaggregates by prevalence tier to expose which configuration wins at each tier rather than on a gap-adjusted average. The pattern reverses across tiers: M1 leads on common categories, M0 on moderate, and B0 on rare. Multi-agent specialisation therefore benefits the clauses models already extract well, not the ones they fail on.[1]

[1]The rare-tier mean masks per-category heterogeneity. M0 leads decisively on specific rare categories where the gate's precision gain compensates the recall cost: *Source Code Escrow* (F_2=1.000 vs. B0 0.943), *Joint IP Ownership* (+0.142 over B0), *No-Solicit of Customers* (+0.233). The tier-mean ranking is dominated by Gemini-family cascade collapse on a subset of rare categories (§5.5), not by uniform M0 weakness across the rare tier.

Table 5.4: Mean F_2 by model compute tier × configuration, under the tier assignment of Table 3.2. Light: Claude Haiku 4.5, Gemini 2.5 Flash, Gemini 3 Flash, GPT-4.1 Mini, GPT-4.1 Nano. Mid: Claude Sonnet 4, Claude Sonnet 4.6, GPT-4.1. Heavy: Claude Opus 4.6, Gemini 2.5 Pro, Gemini 3.1 Pro. Bold: best per row. The architectural ranking inverts across the tier axis: Classify-Extract (M0) wins on Light and Mid, Zero-Shot (B0) wins on Heavy. The F_2 gap between Heavy and Light tiers is 0.106 under B0 and 0.054 under M0 — decomposition narrows the capability gap by 49%. Configuration names defined in Table 3.1.

Tier	B0	B1	M0	M1	M2
Light (5 models)	0.744	0.740	**0.751**	0.743	0.728
Mid (3 models)	0.844	0.848	**0.855**	0.812	0.844
Heavy (3 models)	**0.851**	0.839	0.805	0.844	0.839

5.4 H3 — Architecture vs. Prompt Content (Multi-Agent (Full) vs. Multi-Agent (Prompt-Only))

The Multi-Agent (Full) (M1) router agreed with the deterministic category-to-specialist mapping on 100.00% of requests for ten of eleven models, and on 99.03% for GPT-4.1 Nano (29 mis-routes out of 2,999), yielding 99.91% routing accuracy across the panel. The routed-call distribution across specialists is uneven — `ip_commercial` 11,802 calls, `temporal_renewal` 11,364, `risk_liability` 9,068 — reflecting the per-specialist category populations (17, 11, 13) and CUAD's prevalence-weighted sampling rather than a routing defect. Routing is empirically a deterministic lookup: the router's decisions can be replicated by a static dictionary mapping category names to specialist labels at zero LLM cost. We therefore present no figure for routing accuracy and treat M1 vs. M2 as a test of whether dispatch architecture — with routing held empirically constant — adds value over a single-agent prompt containing the same specialist text.

Table 5.7 reports the M1–M2 contrasts on F_2 and on average Jaccard span overlap.

Despite identical routing, Multi-Agent (Full) (M1) and Multi-Agent (Prompt-Only) (M2) are statistically distinguishable on F_2 for all 11 models after BH correction (McNemar two-sided, α=0.05, m=11; Table 5.7). The direction is inconsistent: M1 outperforms M2 on 5 models, M2 on 6. There is no systematic family pattern. On the secondary axis of span quality, Multi-Agent (Prompt-Only) dominates 10 of 11 models with a +0.071

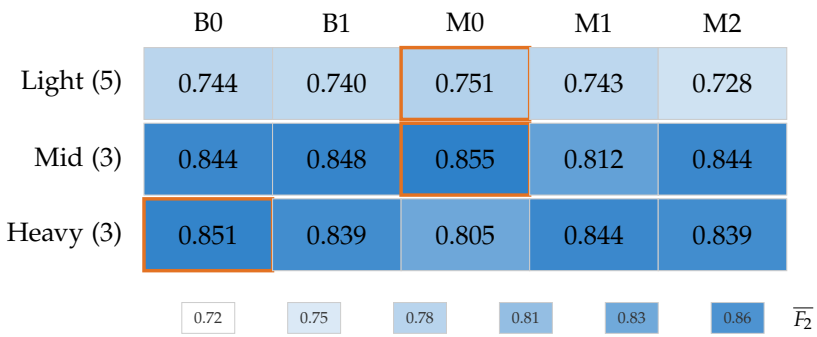


Figure 5.2: Capability tier × configuration heatmap of mean F_2. Cells coloured by $\overline{F_2}$; orange outline marks the best configuration per tier. Classify-Extract (M0) wins on Light and Mid tiers; Zero-Shot (B0) wins on Heavy. Decomposition narrows the Heavy–Light capability gap from 0.106 under B0 to 0.054 under M0 (49% reduction); the same decomposition penalises Heavy models on F_2 because they already absorb long-context load in a single call (see §6.1).

Jaccard advantage, indicating that the consolidated single-agent prompt produces more precise, tighter extractions than the routed specialist pipeline (contribution C4).

H3 verdict. H3 is supported at the per-model level: Multi-Agent (Full) and Multi-Agent (Prompt-Only) are statistically distinguishable on F_2 for all 11 models after BH correction (McNemar two-sided, α=0.05, m=11). The direction of the difference, however, is inconsistent — M1 outperforms M2 on 5 models, M2 on 6 — and pooled F_2 is identical to three decimal places (0.790 vs. 0.790). H3 therefore confirms that dispatch architecture alters model behaviour, not that it improves it. The direction inconsistency is itself a finding: routing overhead hurts precision-oriented models (GPT-family, $\Delta F_2 = -0.056$ to -0.076) and helps recall-deficient ones (Gemini 3 Flash, $\Delta F_2 = +0.138$). On the secondary axis of span quality, M2 dominates 10 of 11 models with a +0.071 Jaccard advantage (contribution C4): the single-agent consolidated prompt produces tighter, more precise extractions than the routed specialist pipeline. A deterministic lookup replaces the LLM router without measurable F_2 loss.

Table 5.5: H1 paired tests: $\Delta F_2 = F_2(\text{multi}) - F_2(\text{B0})$ computed on the sample-ID intersection (so the row ΔF_2 may differ from the full-run difference implied by Table 5.3 when one of the pair is a partial run — Claude Sonnet 4 and GPT-4.1). Paired-bootstrap 95% CIs (B=300, `seed=42`), BH-corrected significance (α=0.05, m=22). ✓: BH-reject H_0. Note: BH significance is decided by the McNemar exact p-value; the bootstrap CI on ΔF_2 is reported separately and may not align with the McNemar verdict on near-zero effects where the discordant-pair distribution and the resampled-mean distribution diverge.

	M0 vs. B0			**M1 vs. B0**		
Model	ΔF_2	CI_{95}	BH	ΔF_2	CI_{95}	BH
Claude Haiku 4.5	+0.009	[−0.009, +0.026]	✓	+0.052	[+0.038, +0.066]	✓
Claude Opus 4.6	+0.001	[−0.011, +0.014]	✓	−0.004	[−0.017, +0.009]	✓
Claude Sonnet 4	+0.025	[+0.016, +0.043]	✓	−0.025	[−0.031, −0.001]	✓
Claude Sonnet 4.6	+0.047	[+0.032, +0.062]	✓	+0.021	[+0.002, +0.038]	✓
Gemini 2.5 Flash	−0.210	[−0.234, −0.181]	✓	−0.020	[−0.036, −0.003]	✓
Gemini 2.5 Pro	−0.121	[−0.143, −0.094]	✓	+0.006	[−0.006, +0.020]	✓
Gemini 3 Flash	+0.037	[+0.015, +0.055]	✓	−0.039	[−0.060, −0.014]	✓
Gemini 3.1 Pro	−0.018	[−0.035, −0.006]	✓	−0.021	[−0.034, −0.007]	×
GPT-4.1	−0.040	[−0.061, −0.026]	✓	−0.060	[−0.083, −0.042]	✓
GPT-4.1 Mini	+0.025	[+0.011, +0.035]	×	−0.060	[−0.081, −0.039]	×
GPT-4.1 Nano	+0.171	[+0.143, +0.195]	✓	+0.059	[+0.025, +0.091]	✓

5.5 Classification-Gate Cascade

Table 5.8 quantifies the M0 cascade. The M0 classifier's `predicted_no_clause` decision bypasses the extractor and records "no related clause" as the final answer. We report the FN-rate ($\text{FN}/(\text{TP}+\text{FN})$) for B0 and M0, and the share of M0 false negatives attributable to a gate rejection rather than a downstream extraction failure.

The cascade is the dominant cause of M0's F_2 regression on the Google family. On Gemini 2.5 Flash, M0 FN-rate jumps to 42.2% from B0's 9.6% — a 4.4× amplification — and 97.9% of those FNs are gate-attributable. The cascade interacts with category prevalence: rare-tier F_2 on Gemini 2.5 Flash drops from 0.719 (B0) to 0.381 (M0), a 0.338-point loss that fully accounts for the rare-gap widening in Section 5.3.

The cascade is a recall–precision exchange, not pure regression. The complement of lost recall is gained precision. Table 5.9 reports ω, the over-extraction rate (Eq. 2.5).

Table 5.6: Per-tier F_2 on common categories (F_2^c) and rare–common gap $g = F_2^c - F_2^r$ for all five configurations. Lower g is better. M0's mean gap (0.284) exceeds B0's (0.276) due to gate-induced rare-tier collapse on Gemini 2.5 Flash/Pro. M1's mean gap (0.342) is worst, driven by GPT-family specialist failure on rare categories. Only B1 and M2 reduce the gap relative to B0 (means 0.240 and 0.226).

	B0		**B1**		**M0**		**M1**		**M2**	
Model	F_2^c	g	F_2^c	g	F_2^c	g	F_2^c	g	F_2^c	g
Claude Haiku 4.5	0.926	0.329	0.917	0.263	0.770	0.074	0.950	0.296	0.938	0.306
Claude Opus 4.6	0.965	0.219	0.950	0.169	0.937	0.202	0.966	0.260	0.953	0.256
Claude Sonnet 4	0.958	0.245	0.955	0.241	0.940	0.211	0.948	0.257	0.937	0.195
Claude Sonnet 4.6	0.947	0.269	0.929	0.149	0.941	0.220	0.928	0.245	0.929	0.175
Gemini 2.5 Flash	0.935	0.216	0.929	0.227	0.725	0.344	0.946	0.277	0.934	0.153
Gemini 2.5 Pro	0.958	0.268	0.960	0.272	0.919	0.515	0.969	0.258	0.954	0.323
Gemini 3 Flash	0.950	0.324	0.798	0.337	0.944	0.355	0.923	0.346	0.751	0.144
Gemini 3.1 Pro	0.966	0.225	0.955	0.236	0.962	0.289	0.963	0.262	0.965	0.189
GPT-4.1	0.950	0.246	0.947	0.220	0.856	0.179	0.964	0.533	0.956	0.219
GPT-4.1 Mini	0.913	0.196	0.947	0.254	0.955	0.233	0.950	0.421	0.936	0.293
GPT-4.1 Nano	0.549	0.296	0.741	0.280	0.708	0.296	0.775	0.608	0.658	0.234
Mean	0.929	0.276	0.912	0.240	0.878	0.284	0.940	0.342	0.919	0.226

Across the panel, M0 cuts mean ω from 0.320 (B0) to 0.130 — a 59% reduction in false positives on negatively-labelled instances. The Gemini family benefits most (ω drops from 0.352 to 0.099) but pays the steepest recall cost. M0 therefore trades recall for precision via the gate. The trade is favourable on F_1 (M0 leads pooled, Table 5.2) and unfavourable on F_2 (B0 leads pooled). Which side of the trade is preferable depends on the deployment cost ratio of missed clauses vs. false extractions, a question revisited in Chapter 6.

Cascade predictability. The cascade severity correlates with the base model's B0 over-extraction rate: the more liberally a model extracts under B0, the more recall it surrenders at the gate. Gemini 2.5 Pro ($\omega_{B0} = 0.456$) loses 0.121 F_2 points to M0; GPT-4.1 Nano ($\omega_{B0} = 0.085$) *gains* 0.171 F_2 points, because the gate corrects its B0 laziness (55.5% FN-rate). A practitioner can predict whether M0 will help or hurt a given model by running a cheap B0 calibration pass and inspecting ω_{B0} against the FN-rate profile. Figure 5.4 visualises the cascade as a movement in the (ω, λ) plane: arrows connect each model's B0 cell to its M0 cell, and the up-and-leftward shift on Gemini Flash/Pro is the cascade.

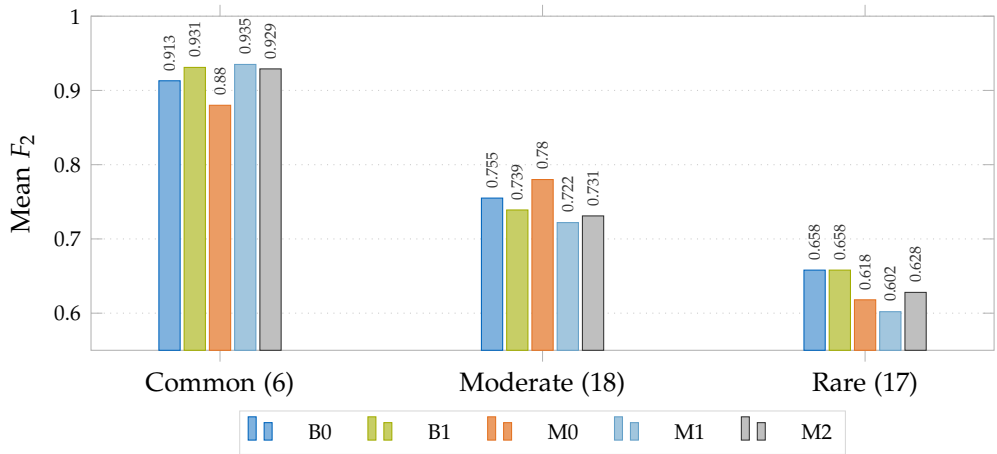


Figure 5.3: Mean F_2 by CUAD prevalence tier $\times$ configuration, pooled across the eleven-model panel. Tier partition derived from positive-instance prevalence in the 102-contract test split (common $>$ 30% positive, moderate 10–30%, rare $<$ 10%); marginally different from the ContractEval F1-based partition (6/14/21) without changing the qualitative ranking. The best configuration per tier differs: M1 wins common (0.935), M0 wins moderate (0.780), B0 and B1 tie on rare (0.658). Multi-agent decomposition helps on categories where the base model is already strong; H2 is rejected because it predicts the opposite direction for rare categories.

5.6 Span Grounding Rate

The grounding rate measures the fraction of an extraction's character span that is reproducible verbatim from the source contract; an ungrounded span is either paraphrased or hallucinated. Two views are reported in Table 5.10: pooled across all sampled instances (denominator includes the $\approx$58% negative-label cases where the model returns "no related clause" and grounding is undefined, conventionally scored 0), and restricted to instances on which the model emitted at least one span.

The predicted-clause column is the audit-relevant denominator: a missed clause has no span to ground, but every emitted span is auditable in principle. Under that view, Multi-Agent (Full)'s structured-output schema (Listing 4.4) guarantees verbatim citation by construction, and Classify-Extract's extractor inherits the same scaffold from the M1 specialist prompts. The two single-call generative configurations Zero-Shot and Multi-Agent (Prompt-Only) leave roughly half of their extractions paraphrased or hallucinated

Table 5.7: H3 contrasts: Multi-Agent (Full) (M1) vs. Multi-Agent (Prompt-Only) (M2). Both configurations are tied on pooled F_2 ($\overline{\Delta F_2} = -0.0002$; 5 M1 wins, 6 M2 wins). On span quality (Jaccard), Multi-Agent (Prompt-Only) dominates on 10 of 11 models, mean advantage +0.071. BH: BH-corrected significance (α=0.05, m=11). All 11 contrasts are significant. The single M1 exception on Jaccard is Gemini 3 Flash, where Multi-Agent (Prompt-Only) collapses (Section 5.5).

Model	F_2(M1)	F_2(M2)	ΔF_2	BH	Jacc. (M1)	Jacc. (M2)
Claude Haiku 4.5	0.835	0.822	+0.013	✓	0.403	**0.449**
Claude Opus 4.6	0.860	0.846	+0.014	✓	0.429	**0.492**
Claude Sonnet 4	0.826	0.842	−0.016	✓	0.491	**0.524**
Claude Sonnet 4.6	0.834	0.836	−0.002	✓	0.397	**0.460**
Gemini 2.5 Flash	0.810	0.832	−0.022	✓	0.384	**0.462**
Gemini 2.5 Pro	0.834	0.816	+0.018	✓	0.375	**0.440**
Gemini 3 Flash	0.747	0.609	+0.138	✓	**0.345**	0.332
Gemini 3.1 Pro	0.840	0.854	−0.014	✓	0.420	**0.534**
GPT-4.1	0.777	0.853	−0.076	✓	0.298	**0.445**
GPT-4.1 Mini	0.777	0.833	−0.056	✓	0.319	**0.387**
GPT-4.1 Nano	0.547	0.545	+0.001	✓	0.272	**0.396**
Mean	0.790	0.790	−0.000	—	0.376	**0.447**

— the dominant failure mode is extraction of *plausible* legal text that cannot be located verbatim in the source contract. This finding qualifies contribution C4: the auditability advantage of multi-agent topologies is not solely the trace structure (Section 4.6) but also the schema-enforced grounding that the topology makes operationally feasible.

5.7 Cost-Efficiency Frontier

Table 5.11 reports per-run provider costs and mean ratios relative to B0.

Figure 5.6 plots all 55 cells in the (F_2, cost) plane. B0 occupies the Pareto frontier at every cost level at which it appears. The three Gemini M0 cells are strictly Pareto-dominated: lower F_2 *and* higher cost than B0 Gemini alternatives.

The 23–67% mean cost premium attached to multi-agent configurations is not compensated by aggregate F_2 gains. In workflows where per-agent traces add operational value

Table 5.8: M0 classification-gate rejection cascade. FN-rate = FN/(TP + FN). Gate-FN: false negatives where the classifier emitted `predicted_no_clause`. Gate % = Gate-FN / M0 FN. On average 82.7% of M0's false negatives originate at the gate; Gemini 2.5 Flash and Pro lose ≥97.8% of their recall to upstream rejection.

Model	B0 FN-rate	M0 FN-rate	M0 FN	Gate-FN	Gate %
Claude Haiku 4.5	13.4%	20.6%	255	210	82.4%
Claude Opus 4.6	6.9%	11.5%	142	95	66.9%
Claude Sonnet 4	7.9%	9.9%	123	87	70.7%
Claude Sonnet 4.6	10.9%	12.6%	156	131	84.0%
Gemini 2.5 Flash	9.6%	42.2%	520	509	97.9%
Gemini 2.5 Pro	7.9%	32.6%	403	394	97.8%
Gemini 3 Flash	17.2%	17.9%	221	197	89.1%
Gemini 3.1 Pro	9.9%	15.5%	191	169	88.5%
GPT-4.1	9.2%	16.6%	207	191	92.3%
GPT-4.1 Mini	13.4%	11.8%	147	103	70.1%
GPT-4.1 Nano	55.5%	36.3%	452	318	70.4%
Mean	14.7%	20.7%	256	218	82.7%

(obligation tracking, audit-trail generation), this premium may be acceptable; Chapter 6 elaborates these conditions.

5.8 Contract-Length Moderation

The cascade analysis (Section 5.5) and tier analysis (Figure 5.3) treat all contracts uniformly. Contract length provides an orthogonal moderating variable: the context-window motivation for multi-agent decomposition predicts that routing clauses to specialists should help on *longer* contracts, where a monolithic prompt must attend over more irrelevant text. We test this prediction by splitting all 102 CUAD test contracts into quartiles by character count and computing per-configuration F_2 within each quartile. The main evaluation already uses all 102 contracts of the CUAD test split, so the quartile analysis here is a length-stratified view of the headline results in Sections 5.2–5.5 rather than a separate evaluation envelope.

Table 5.12 shows the opposite of the predicted pattern. On Q1 contracts (under 14,384 characters), M1 leads by +3.7 percentage points over B0 and M0 by +2.0 points. On

Table 5.9: Over-extraction rate ω (Eq. 2.5): fraction of negative-label instances on which the model emits a non-empty span. Bold: best per row. The classification gate cuts ω by more than half on average; the Gemini family gains the most precision at the recall cost documented in Table 5.8.

Model	ω(B0)	ω(B1)	ω(M0)	ω(M1)	ω(M2)
Claude Haiku 4.5	.478	.369	**.158**	.191	.316
Claude Opus 4.6	.330	.436	**.169**	.184	.335
Claude Sonnet 4	.335	.400	**.169**	.248	.236
Claude Sonnet 4.6	.423	.538	**.150**	.198	.433
Gemini 2.5 Flash	.392	.422	**.082**	.182	.354
Gemini 2.5 Pro	.456	.604	**.076**	.303	.454
Gemini 3 Flash	.321	.330	**.120**	.135	.450
Gemini 3.1 Pro	.236	.242	**.116**	.199	.242
GPT-4.1	.221	.459	**.102**	.119	.335
GPT-4.1 Mini	.223	.455	.174	**.155**	.409
GPT-4.1 Nano	**.085**	.181	.137	.057	.175
Mean	.320	.385	**.130**	.179	.339

Q4 contracts (over 62,272 characters), B0 leads by +3.2 points over M0 and +4.1 over M1. The crossing point lies between Q2 and Q3 (≈25,900 characters). The mechanism is plausibly gate-induced: on long contracts the M0 classifier is exposed to larger token windows, increasing the probability of conservative `predicted_no_clause` decisions that trigger the cascade documented in Section 5.5. For M1, long-contract degradation may reflect specialist prompt dilution when many clause types from one domain appear in a single long document. In either case, multi-agent decomposition as implemented offers no context-window advantage over a zero-shot prompt on CUAD-class contract lengths.

5.9 Answers to Research Questions

RQ1 (which architectural pattern wins, and under what conditions). No architectural pattern wins universally on F_2. Pooled F_2 ranks within a 0.011-point band, and per-model winners are split four ways across the panel. The dominant signal is the model-family interaction: the Anthropic family peaks at B1 ($\overline{F_2} = 0.851$), the Google family at B0 (0.826), and the OpenAI family at M0 (0.783). The choice of evaluation metric also flips the ranking: M0 dominates the panel on F_1 (0.798 vs. 0.737 for B0) by

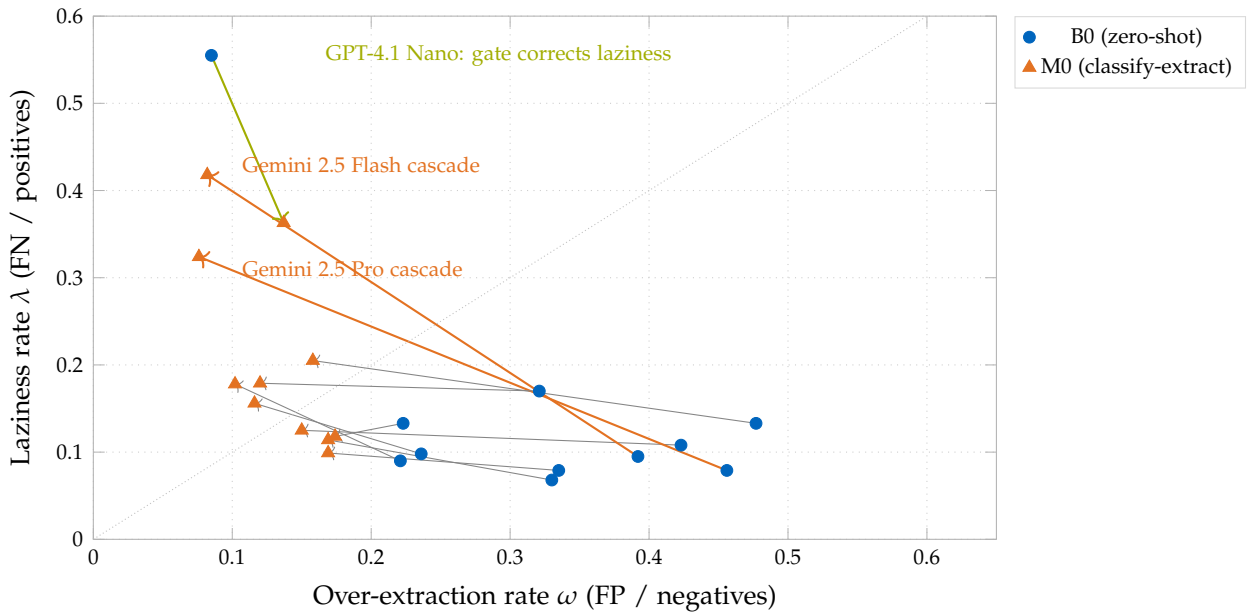


Figure 5.4: Recall–precision exchange under Classify-Extract. Each B0 model (•) maps to its M0 counterpart (▲); arrows show the per-model trajectory in the (ω, λ) plane. Most models move *left and slightly up* — a moderate precision gain (lower ω) paid with a small recall loss (higher λ). On Gemini 2.5 Flash and Pro (orange arrows) the recall loss is severe: λ jumps to > 0.32, the cascade documented in Table 5.8. On GPT-4.1 Nano (green arrow) the trajectory reverses: the gate corrects the model's high B0 laziness (λ=0.555 B0 $\rightarrow$ 0.363 M0). The dotted diagonal marks $\omega = \lambda$ as an orientation reference.

suppressing over-extraction. The condition under which a multi-agent pipeline (M0) is preferable on F_1 is therefore: the model is OpenAI-family, the practitioner accepts the 1.67× cost premium, and precision is the primary criterion. Under the pre-registered primary metric F_2, no multi-agent configuration consistently dominates B0; all other conditions favour simpler baselines.

RQ2 (cost-effectiveness vs. single-agent baselines). B0 occupies the Pareto frontier at every cost level. Multi-agent configurations carry mean cost premiums of 1.23× (M1) to 1.67× (M0) without compensating F_2 gains pooled across the panel. The only cost premium that partially pays back is M0 on the OpenAI family (+0.052 F_2 over B0), but even that is dominated in F_2/\$ by cheaper B0 deployments on the Google family.

Table 5.10: Mean grounding rate by configuration. *All*: pooled over all ≈2,999 samples per cell, including "no related clause" returns scored 0. *Predicted-clause only*: conditioned on a non-empty prediction. Multi-Agent (Full) (M1) achieves 1.000 pooled across every model in the predicted-clause column: the structured specialist output enforces contract-text citation. Classify-Extract (M0) reaches 0.893. Zero-Shot (B0) and Multi-Agent (Prompt-Only) (M2), which return free-text extractions, leave 44–58% of predicted spans ungrounded.

Config	**B0**	**B1**	**M0**	**M1**	**M2**
All samples	0.299	0.373	0.375	**0.456**	0.248
Predicted-clause only	0.543	0.594	0.893	**1.000**	0.415

Cost-aware deployment of clause extraction at CUAD-class scale should default to single-agent zero-shot on a small, fast model unless the deployment context provides specific reasons to prefer higher precision over recall.

RQ3 (auditable reasoning traces). Addressed by system design in Chapter 4. The M1 trace structure produces explicit parent-child decision attribution — routing decision, specialist extraction, validation outcome — that B0 does not. Three tradeoffs attend this auditability surplus. Token storage scales with trace depth: each M1 call emits a LangFuse trace of approximately 4× the token volume of a B0 call (orchestrator hop plus three specialist sub-prompts logged at inference time), raising persistent storage requirements proportionally. Attribution faithfulness is bounded by the trace format: LangFuse records the *submitted* prompt and *returned* completion but cannot verify that the model's stated reasoning reflects its actual inference process, so the trace is auditable but not interpretable in a mechanistic sense. Routing redundancy is overhead without decision diversity: the M1 router is empirically a deterministic lookup (Section 5.4, 99.91% accuracy), so the orchestrator adds latency and cost without altering routing outcomes. Whether the auditability surplus justifies these costs is a deployment question revisited in Chapter 6.

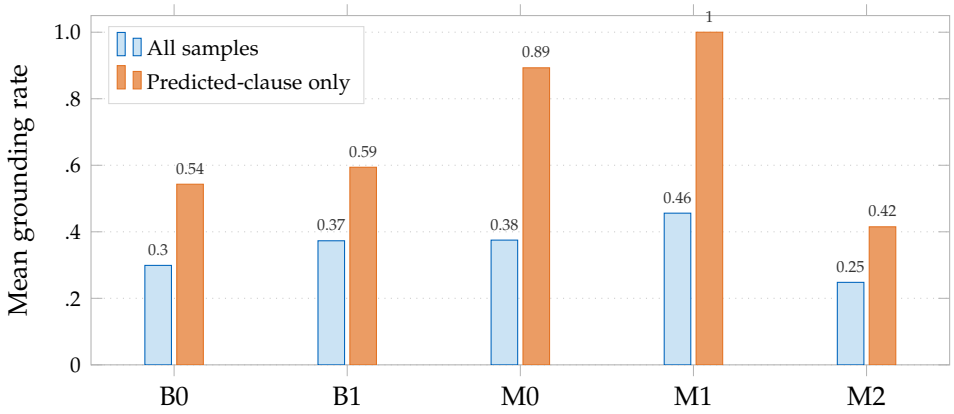


Figure 5.5: Mean grounding rate by configuration, two denominators. *All samples* (blue) includes the ≈58% "no related clause" returns where grounding is undefined and conventionally scored 0; *predicted-clause only* (orange) is the audit-relevant view, conditioned on a non-empty extraction. Multi-Agent (Full) (M1) reaches 1.000 verbatim citation on every predicted clause across all eleven models; Classify-Extract (M0) reaches 0.893. The two free-text generative configurations (Zero-Shot, Multi-Agent (Prompt-Only)) leave 46–58% of their predicted spans paraphrased or hallucinated.

Table 5.11: Provider cost per run (≈2,999 samples) and per-model mean ratio relative to B0. M0's 1.67× premium reflects the additional classifier call; M1's 1.23× reflects the orchestrator hop.

Model	B0 ($)	B1 ($)	M0 ($)	M1 ($)	M2 ($)
Claude Haiku 4.5	36.70	43.11	60.66	45.00	43.29
Claude Opus 4.6	185.02	213.30	302.96	215.49	211.10
Claude Sonnet 4	94.04	125.98	184.58	127.17	125.41
Claude Sonnet 4.6	111.44	129.59	184.11	137.75	128.24
Gemini 2.5 Flash	10.64	14.72	13.73	13.31	13.99
Gemini 2.5 Pro	43.39	61.77	58.69	55.57	58.21
Gemini 3 Flash	16.55	19.09	26.24	19.65	19.18
Gemini 3.1 Pro	65.75	72.46	104.57	78.73	72.69
GPT-4.1	38.07	77.88	91.50	56.11	73.68
GPT-4.1 Mini	13.17	15.78	21.42	13.22	15.78
GPT-4.1 Nano	3.17	3.64	5.10	3.64	3.63
Mean ratio	1.00×	1.30×	1.67×	1.23×	1.27×

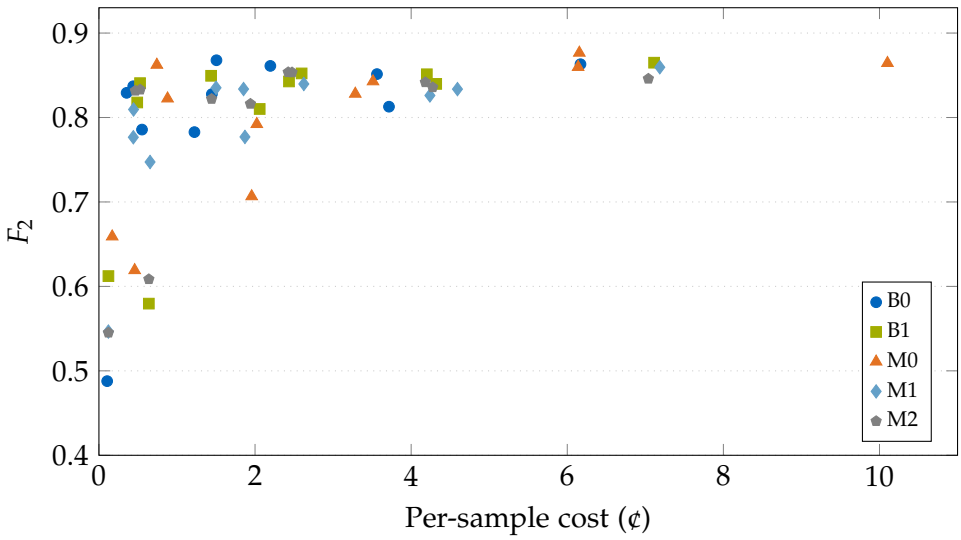


Figure 5.6: F_2 vs. per-sample cost across all 55 cells. The Pareto frontier is occupied by B0 at every cost level. The three Gemini M0 cells (Gemini 2.5 Flash, Gemini 2.5 Pro, and Gemini 3 Flash) are strictly Pareto-dominated: lower F_2 *and* higher cost than B0 alternatives.

Table 5.12: Mean F_2 by contract-length quartile (102 contracts, ≈26 per quartile), pooled across all eleven models. Quartile boundaries: 14,384 / 25,922 / 62,272 characters. Multi-agent configurations (M0, M1) outperform B0 on short contracts (Q1–Q2) by +1.7 to +3.7 percentage points and underperform on long contracts (Q3–Q4) by −0.5 to −4.1 percentage points. The pattern is opposite to the context-window motivation.

Quartile	Range (chars)	B0	B1	M0	M1	Best
Q1 (shortest)	0–14,384	0.756	0.767	**0.776**	**0.793**	M1
Q2	14,385–25,922	0.791	0.803	**0.808**	0.800	M0
Q3	25,923–62,272	0.806	0.793	0.801	0.799	B0
Q4 (longest)	>62,272	**0.849**	0.819	0.817	0.808	B0

6 Discussion

No multi-agent configuration improves on the strongest single-agent baseline by more than 0.011 pooled F_2 points across eleven models.

6.1 The Headline Finding and Its Conditions

On the 102-contract CUAD test split under the upsampled-positive class distribution described in §3.2, no multi-agent configuration improves on the strongest single-agent baseline by more than 0.011 points of mean F_2 pooled across the eleven-model panel. Under the pre-registered decision rule of Section 3.5, H1 is rejected in its universal form: no multi-agent configuration achieves consistent superiority across the eleven-model panel, and eleven of the 22 per-model contrasts are negative or zero. Conditional improvement exists for individual (model, configuration) cells — Classify-Extract (M0) on the OpenAI family and lighter Anthropic models — but the pooled signal is dominated by the model-family interaction. H2 is rejected: both pipeline decomposition (Classify-Extract, M0) and specialist routing (Multi-Agent (Full), M1) widen the rare–common F_2 gap relative to Zero-Shot (B0). H3 is supported per model — Multi-Agent (Full) (M1) and Multi-Agent (Prompt-Only) (M2) are statistically distinguishable on every model after BH correction — but the direction is inconsistent across the panel and the pooled F_2 difference is zero to three decimals.

The single most informative pattern is the model-family interaction. The Anthropic family achieves its peak under CoT (B1, $\overline{F_2}$=0.851); the Google family under Zero-Shot (B0, 0.826); the OpenAI family under Classify-Extract (M0, 0.783). The capability-tier gradient in Table 5.4 is consistent with this: light- and mid-tier models gain from M0's calibration, while heavy-tier models are better served by B0, with the F_2 capability gap between Heavy and Light tiers narrowing from 0.106 under B0 to 0.054 under M0 — a 49% reduction. Decomposition acts partly as a capability-equalising effect, but at a cost (Section 6.2) that is not recovered for heavy models that already absorb long-context load in a single call. Prior work has observed that a well-prompted single agent matches multi-agent decomposition under stronger base models [Gao+25]; the

present data reproduce this: the Anthropic Opus tier achieves its best F_2 under B1, not under any decomposed architecture.

The implication for the open empirical question of when multi-agent decomposition is justified is that, on this benchmark, the conditions are narrower than the literature on agentic legal AI implies. A practitioner asking whether decomposition helps a CUAD-class extraction task must answer two prior questions: which model family, and which capability tier. On the eleven models tested, only the OpenAI family receives a meaningful multi-agent benefit at the pooled level, and even that benefit is dominated in F_2/\$ by a B0 deployment on a cheaper Google Flash model.

6.2 The Cascade as a Recall–Precision Exchange

Section 5.5 identifies the classification-gate cascade as the dominant failure mode of Classify-Extract (M0) on the Google family: 97.8–97.9% of M0 false negatives on Gemini 2.5 Flash and Pro originate at the gate rather than at the extractor. The discussion here defends the framing of this as a recall–precision exchange rather than as pure regression.

The classifier reduces over-extraction substantially: mean ω drops from 0.320 under Zero-Shot (B0) to 0.130 under Classify-Extract (M0), a 59% reduction (Table 5.9). Under F_1, which weights precision and recall equally, the gate's gain is sufficient to push M0 to the highest pooled score ($\overline{F_1}$=0.798 vs. B0's 0.737). Under F_2, the gain is insufficient to compensate the recall loss. The cascade is thus not evidence that M0 is broken, but evidence that M0 makes a different operating-point trade. Under the pre-registered primary metric F_2, the trade is unfavourable for 7 of 11 models. The practitioner question is which operating point matches the deployment cost structure.

The cascade direction is predictable from a single-agent calibration pass. Models with high B0 over-extraction rates (Gemini 2.5 Pro ω_{B0}=0.456; Gemini 2.5 Flash 0.392) inherit a permissive prior on negative instances; the gate amplifies the cautious prior, suppressing recall. Models with low B0 over-extraction (GPT-4.1 Nano ω_{B0}=0.085) gain from the gate because their single-agent extraction loses ground to laziness rather than to over-extraction. The practitioner heuristic is: run B0, inspect ω_{B0}, and deploy M0 only if ω_{B0} is moderate-to-low; within the present panel, within the present eleven-model panel, models with $\omega_{B0} \lesssim 0.20$ gain on F_1 under M0, those above that threshold do not; this boundary is indicative, not validated.

This result connects to the failure-mode taxonomy of Cemri, Pan, Yang, et al. [Cem+25], who identify the inter-agent verification gap as a leading cause of multi-agent regression: the downstream extractor cannot override the upstream classifier's silent rejection. A reflective verifier agent — absent in M0 — would be the natural mitigation, and is identified as an untested architectural pattern in Section 7.3.

6.3 Routing as a Deterministic Lookup

The Multi-Agent (Full) (M1) router agreed with a static category-to-specialist mapping on 99.91% of requests across the panel (Section 5.4). The natural reading is that routing in this design space reduces to a table lookup keyed on the category name, a negative finding for the LLM-router subspace illustrated by ChatLaw v2 [Cui+24] and L-MARS [WY25]. Two qualifications are worth stating.

First, the result depends on three architectural commitments that do not hold in every multi-agent design: the specialist domains were disjoint and descriptively named; the categories carried indicators the router could read directly from the category label; and the partition was fixed before any model was run. In designs where domains overlap, where the category vocabulary is open-ended, or where the routing decision must integrate retrieved context, the router's work becomes non-trivial and routing accuracy falls below ceiling. The generalisable claim is therefore narrower: in a closed-vocabulary specialist-routing design, the router's contribution is replicable by a deterministic dictionary at zero LLM cost.

Second, Multi-Agent (Full) (M1) and Multi-Agent (Prompt-Only) (M2) differ statistically on every model despite identical routing. The difference is not in dispatch but in prompt format: M1 sees one focused specialist preamble per call, M2 sees the textual union of three. Consolidating the specialist content into a single prompt produces measurably better span coverage (Jaccard $+0.071$ pooled, Table 5.7) without changing F_2. The mechanism is plausibly attentional: the union prompt presents all three category-cluster framings simultaneously and lets the model choose, rather than committing the dispatch decision before the contract is read.

The combined implication for the design of legal-AI multi-agent systems is that the value of specialist decomposition lies in the prompt content, not in the dispatch architecture. A practitioner with a fixed specialist taxonomy is better served by a single agent receiving the union of specialist instructions than by a router-plus-specialists topology, and eliminates the 23% orchestrator cost premium.

6.4 The β=2 Choice and Operating-Point Sensitivity

The pooled architectural ranking depends on the chosen β. At β=1 the ranking is Classify-Extract > Multi-Agent (Full) > Zero-Shot > Multi-Agent (Prompt-Only) > CoT. At β=2 the ranking is Zero-Shot > CoT > Classify-Extract > Multi-Agent (Full) $\approx$ Multi-Agent (Prompt-Only). The architectural conclusion of the thesis therefore depends, in part, on the β choice.

The commitment to β=2 is justified in Section 3.4 by the deployment cost structure of CLM review: a missed clause carries the legal cost of an undisclosed obligation, whereas a spurious extraction is corrected at the lower cost of attorney review time. The empirical question of where the cost ratio actually sits is deployment-dependent and falls outside the scope of this thesis. The sensitivity is mitigated by reporting F_1, recall, precision, ω, and FN-rate alongside F_2, so a deployment with a different cost ratio can re-rank architectures from the same data. The practitioner recommendation is conditional on β: at β=2 deploy B0; at β=1 deploy M0; below β=1 (where false positives are costlier than false negatives, as in some compliance-screening workflows), M0's calibration advantage widens further.

6.5 Benchmark-to-Production Gaps

The CUAD-based evaluation establishes what current architectures achieve under idealised conditions: clean born-digital text, a fixed forty-one-category taxonomy, expert-annotated ground truth, and a single contract per query.

1. Document quality is the first bottleneck. CUAD contracts are EDGAR filings: clean, machine-readable, consistently formatted. Enterprise repositories contain scanned PDFs of variable OCR quality, handwritten margin annotations, partial digitisations of legacy contracts, and redacted sections. In the authors' assessment of production CLM deployments, document preprocessing quality sets the primary performance ceiling — a factor upstream of model selection. A model achieving high F_2 on CUAD text need not retain that performance on a poorly scanned contract with OCR artefacts. This shifts the practitioner-relevant axis of architectural design from which prompt structure handles the contract to which preprocessing pipeline is needed before any extraction architecture is invoked. The multi-agent design space examined in this thesis is downstream of that question.

2. The task is multi-contract, not single-contract. CUAD evaluates a one-to-one mapping: one contract, one category. The practitioner workflow is retrieve-then-extract: a question ("which supplier contracts expose us to uncapped liability?") is first resolved against a repository of hundreds or thousands of contracts, then the relevant subset is extracted from. This pipeline introduces three challenges that no current public benchmark addresses: cross-contract consistency, deduplication, and result aggregation, and it shifts the architecture's value proposition from extraction accuracy to retrieval-extraction integration. Multi-agent designs in which a retrieval agent precedes an extraction agent are a natural fit for this workflow, and the M0 classify-then-extract pattern is the simplest non-trivial member of that family. The cascade failure mode of Section 6.2 therefore has direct deployment relevance: the same upstream-rejection dynamics that suppress M0 recall on Gemini 2.5 will plausibly suppress retrieval-extraction recall in production. A controlled retrieve-then-extract benchmark on a multi-contract corpus is identified as future work in Chapter 7.

3. Clause taxonomies are domain-specific and open-ended. The forty-one CUAD categories cover general commercial provisions. Construction contracts contain liquidated damages tied to project milestones, variation orders, retention terms, and defects-liability periods; SaaS agreements contain SLA penalties, data-processing addenda, and sub-processor notification clauses. None of these map cleanly to CUAD's category schema. A production CLM system handles an open-ended and evolving taxonomy. The specialist-decomposition design tested here (M1, M2) is a fixed three-domain partition; its extensibility to an open taxonomy is the question that matters for production, and is not what this thesis evaluates.

4. Production has no ground truth. CUAD provides expert annotations against which F_2, Jaccard, and ω can be computed. A production deployment has none. Evaluation proceeds through user feedback, spot-checking, and exception triage — none of which benchmark metrics capture. This is the deployment-side argument for the observability work in Chapter 4: in the absence of ground truth, the most valuable architectural property is whether each decision can be attributed to a specific agent, prompt, and routing branch, so that exceptions can be triaged at the source (Section 4.6). The M0 classifier verdict and the M1 routed-specialist label are precisely the trace primitives a production reviewer needs, even when the underlying F_2 is dominated by a simpler single-agent baseline. The thesis's headline result — that multi-agent F_2 does not beat single-agent — does not therefore translate into "deploy single-agent in production",

because the production utility function includes auditability, not only F_2 (see Section 4.6 and Table 5.7 for the trace-completeness evidence).

5. Cross-contract reasoning is required and unmeasured. Production CLM questions span contracts: "are our liability caps consistent across all vendor agreements?"; "which contracts are affected if Vendor X is acquired?". Cross-contract reasoning requires maintaining context across documents and detecting relationships between clauses in different agreements. Neither CUAD nor the architectures evaluated here address this. Hierarchical multi-agent designs — a portfolio-level orchestrator over per-contract extraction agents — are a natural fit for this workflow and are identified as future work in Section 7.3.

6. Integration determines deployment value. A system with high benchmark accuracy fails in production if it cannot surface results inside the platform of record, in formats downstream systems consume (risk reports, obligation trackers, renewal calendars), with sufficient context to act upon. The post-extraction integration challenge — transforming an extracted clause into an actionable insight — is largely orthogonal to architectural choice within the extraction layer and is the axis on which production CLM platforms compete most directly.

These gaps locate the CUAD evaluation rather than invalidating it: the thesis measures one axis (extraction accuracy under fixed taxonomy and clean text) that is necessary but not sufficient for production deployment. The practitioner argument for multi-agent architectures therefore shifts axis: not that decomposition raises extraction accuracy — the data indicate it generally does not on this benchmark — but that decomposition provides trace primitives (structured per-agent decision logs), modular update boundaries, and integration seams (interface boundaries where components can be replaced or extended) that production deployments need on axes the benchmark does not measure. This is a defensible reason to build multi-agent CLM systems even when the benchmark F_2 ranking favours single-agent baselines. It is a weaker reason than "decomposition wins on accuracy", and the empirical record is better served by stating it cleanly.

6.6 Threats to Validity

The threats formalised in Section 3.7 bound the generalisation of these findings. Three are most consequential.

Single benchmark. The evaluation is on the 102-contract CUAD test split under the upsampled-positive class distribution (§3.2). The findings may not extrapolate to non-CUAD clause taxonomies, to non-English contracts, or to non-U.S. jurisdictions. The contract-length moderation result (Section 5.8, Table 5.12) shows that the architectural ranking is not stable across document lengths within CUAD itself: multi-agent helps on short contracts and hurts on long ones. The architectural ranking observed here is therefore a ranking on this evaluation envelope, not a universal ranking.

Pinned model snapshots. Model identifiers are pinned to specific snapshots in the run metadata. Provider behaviour can shift under silent model updates; the family interaction observed here may not be stable across future provider releases. The qualitative pattern — that architecture choice interacts with model family more than with any within-architecture variable — is consistent with findings reported across task domains in the recent literature [Gao+25; Cem+25], which suggests the pattern is consistent across task domains even if specific point estimates are not.

Closed proprietary panel. Open-weights models are excluded (Section 3.3). The proprietary–open-source gap reported by S. Liu, Z. Li, Ma, et al. [Liu+25] is not re-examined here. The finding that multi-agent decomposition does not improve pooled F_2 may not transfer to a panel of weaker open-weights models, where decomposition's capability-equalisation effect (Section 6.1) might be larger. Open-weights extension is identified in Section 7.3 as the most consequential extension of the present work.

7 Conclusion and Outlook

Model capability tier dominates architectural choice: at this task, investing in a stronger model yields larger F_2 gains than switching from a single-agent to a multi-agent pipeline.

7.1 Summary of Contributions

This thesis evaluated five architectural configurations against eleven frontier proprietary LLMs on the 102-contract CUAD test split, producing 163,644 paired sample-level predictions analysed under a controlled statistical pipeline. The four contributions stated in Chapter 1 are restated here with pointers to the evidence that supports them.

C1 — Cross-architecture benchmark. A cross-family cross-architecture comparison on CUAD at this scale: five architectural configurations, eleven frontier models, 55 official runs, analysed with paired McNemar exact tests, paired bootstrap confidence intervals, and Benjamini–Hochberg correction. The benchmark, prompt registry, run artefacts, and statistical pipeline are released alongside the thesis (Section 4.7).

C2 — Capability-tier gradient. Architectural ranking is not constant across the model panel. Pooled F_2 across the five configurations spans a 0.011-point band and per-model winners are split four ways (Table 5.3). The capability-tier breakdown (Table 5.4) reveals the gradient: light- and mid-tier models gain from Classify-Extract's (M0) calibration; heavy-tier models are dominated by Zero-Shot (B0). No architecture is universally optimal; the choice is conditional on model family, capability tier, and the chosen β of the F_β metric.

C3 — Classification-gate cascade. The pipeline-decomposition pattern (Classify-Extract, M0) suffers a recall–precision exchange mediated by the classification gate. On Gemini 2.5 Flash and Pro, 97.8–97.9% of M0 false negatives originate at the gate rather than at the extractor (Table 5.8). The exchange's direction is predictable from a single-agent calibration pass: models with high Zero-Shot (B0) over-extraction lose recall to the gate; models with low B0 over-extraction gain F_1 from it. Section 6.2 formalises the heuristic and connects it to the inter-agent verification gap taxonomy of Cemri, Pan, Yang, et al. [Cem+25].

C4 — Routing redundancy with span-quality and grounding tradeoffs. The LLM router agreed with a deterministic category-to-specialist dictionary on 99.91% of requests across the panel and on 100% for ten of eleven models. Multi-Agent (Prompt-Only) (M2) matches Multi-Agent (Full) (M1) on F_2 pooled to three decimals and produces tighter spans (Jaccard $+0.071$ on ten of eleven models; Table 5.7). The architectural advantage that does survive ablation is grounding: Multi-Agent (Full)'s schema-enforced specialist output achieves 100% verbatim citation of contract text on every predicted clause, against 54% for Zero-Shot and 42% for Multi-Agent (Prompt-Only) (Table 5.10). The LLM-mediated routing call adds orchestrator cost without measurable F_2 benefit, but the specialist scaffold to which the router dispatches carries the auditability surplus.

7.2 Answers to the Research Questions

Main RQ. *Can a multi-agent framework improve contract clause extraction accuracy beyond single-agent baselines while providing superior explainability?* On accuracy (F_2), multi-agent decomposition is dominated at the pooled level by Zero-Shot (B0), and conditionally improves only on the OpenAI family and on lighter capability tiers. On efficiency (cost-adjusted F_2), multi-agent configurations are Pareto-dominated by B0 deployments on cheaper models (Figure 5.6). On explainability, the pipeline (Classify-Extract, M0) and routed-specialist (Multi-Agent (Full), M1) topologies produce the trace primitives that production review needs (Section 4.6), which is the axis on which the practitioner argument for multi-agent architectures rests (Section 6.5).

RQ1 — Architectural pattern by condition. No pattern wins universally. Pooled F_2 ranks Zero-Shot $>$ CoT $>$ Classify-Extract $>$ Multi-Agent (Full) $\approx$ Multi-Agent (Prompt-Only) within a 0.011-point band. The model-family interaction is the dominant signal:

the Anthropic family peaks under CoT (B1), the Google family under Zero-Shot (B0), and the OpenAI family under Classify-Extract (M0). The prevalence-tier interaction is also large: M1 specialists lead on common-tier categories, M0 on moderate, and B0 on rare (Figure 5.3). The capability-tier interaction completes the picture: Classify-Extract helps light and mid models, Zero-Shot helps heavy models. The answer to RQ1's "under what conditions" clause is therefore: M0 is preferable when the model is OpenAI-family or light-tier, F_1 is the primary metric, and the practitioner accepts the 1.67× cost premium.

RQ2 — Multi-agent vs. single-agent on accuracy and cost. Multi-agent configurations carry mean cost premiums of 1.23× (Multi-Agent (Full), M1) to 1.67× (Classify-Extract, M0) over Zero-Shot (B0) without compensating F_2 gains pooled across the panel. Zero-Shot occupies the Pareto frontier at every cost level (Figure 5.6). The only multi-agent cost premium that pays back in F_2 terms is M0 on the OpenAI family, and even that is dominated in F_2/\$ by a B0 deployment on Gemini 2.5 Flash. On F_1, Classify-Extract leads pooled and the premium is partially justified for precision-critical deployments.

RQ3 — Auditable reasoning traces. The parent–child trace structure of Classify-Extract (M0) and Multi-Agent (Full) (M1) produces decision attribution — routing decision, specialist extraction, validation outcome — that Zero-Shot (B0) cannot (Section 4.6). The trace structure is paired with schema-enforced grounding: Multi-Agent (Full) reaches 100% verbatim contract-text citation on every predicted clause and Classify-Extract reaches 89.3%, against 54.3% for Zero-Shot and 41.5% for Multi-Agent (Prompt-Only) (Table 5.10). The practitioner-relevant tradeoffs are storage cost (approximately 4× the token volume of a B0 call per M1 run), attribution faithfulness (the trace is faithful to the call graph but does not certify the model's internal computation), and routing redundancy (the 99.91% deterministic routing result means the orchestrator layer adds latency without contributing decision diversity).

7.3 Outlook

Three axes most clearly under-determine the conclusion: untested architectural patterns, evaluation beyond CUAD, and cost mechanisms not exercised in the experiments.

Untested architectural patterns. The five-configuration grid covers single-agent, pipeline, and routed-specialist patterns under static and LLM-mediated routing. Three patterns we expect to perform measurably differently remain unevaluated. *Reflective verifier agents* [Shi+23] could absorb the cascade failure mode of Section 6.2 by allowing the extractor to override a silent classifier rejection on confidence-flagged borderline cases. *Tool-calling specialists* that retrieve clause exemplars or jurisdictional indicators before extracting are the natural extension to open-taxonomy production deployments (Section 6.5, Gap 3). *Hierarchical orchestration* — a portfolio-level orchestrator over per-contract extraction agents — targets cross-contract reasoning (Gap 5) and is the architecture we expect to dominate on multi-contract benchmarks that do not yet exist.

Beyond CUAD. Three extensions of the evaluation envelope are immediate. *Open-weights models* are excluded from the present panel; the proprietary–open-source gap reported by S. Liu, Z. Li, Ma, et al. [Liu+25] is the most consequential variable left unexamined, and the capability-equalisation effect of M0 may be larger on weaker base models, in which case the architectural ranking observed here inverts. *Retrieve-then-extract* on a multi-contract corpus is the production-relevant generalisation of M0 and is currently unbenchmarked at the scale required (Section 6.5, Gap 2). *Document-quality stress tests* — OCR noise, mixed-format documents, redactions — would establish whether the architectural ranking is stable under the preprocessing-degraded inputs that dominate enterprise deployments (Gap 1). *Formal user studies* with practising contract reviewers would close the loop on RQ3 by measuring whether the trace primitives produced by M0 and M1 actually shorten review time or improve exception-triage accuracy in production, a question this thesis treats as a system property but does not measure on human reviewers.

Cost mechanisms. Two cost mechanisms not exercised here would change the cost picture (Section 5.7) materially. *Prompt caching*, supported in production by the major providers in the panel, would substantially reduce the marginal cost of per-category re-prompting that dominates the B0 cost structure on long contracts; the 23–67% multi-agent cost premium would shrink correspondingly and the Pareto frontier would re-form. *Structured-output constraints* would reduce M0 classifier-call latency to the point where the cascade exchange is acceptable on more models. The Pareto analysis as given is therefore a snapshot under uniform per-call billing; the deployment-relevant cost ranking under caching and structured-output decoding is open.

The thesis's closing position is that the multi-agent design space for CLM clause extraction is large, that the slice tested here is the slice production teams are most likely to build first, and that this slice does not on its own justify multi-agent deployment over a strong single-agent baseline on accuracy grounds. The case for multi-agent architectures in production rests on axes the benchmark does not measure: auditability, modular update boundaries, integration seams, open-taxonomy extensibility, and cross-contract reasoning. Future work should evaluate those axes directly, rather than continue to test multi-agent decomposition against benchmarks where single-agent long-context already absorbs the workload.

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